

JSS 2 LIVESTOCK FARMING LESSON NOTES

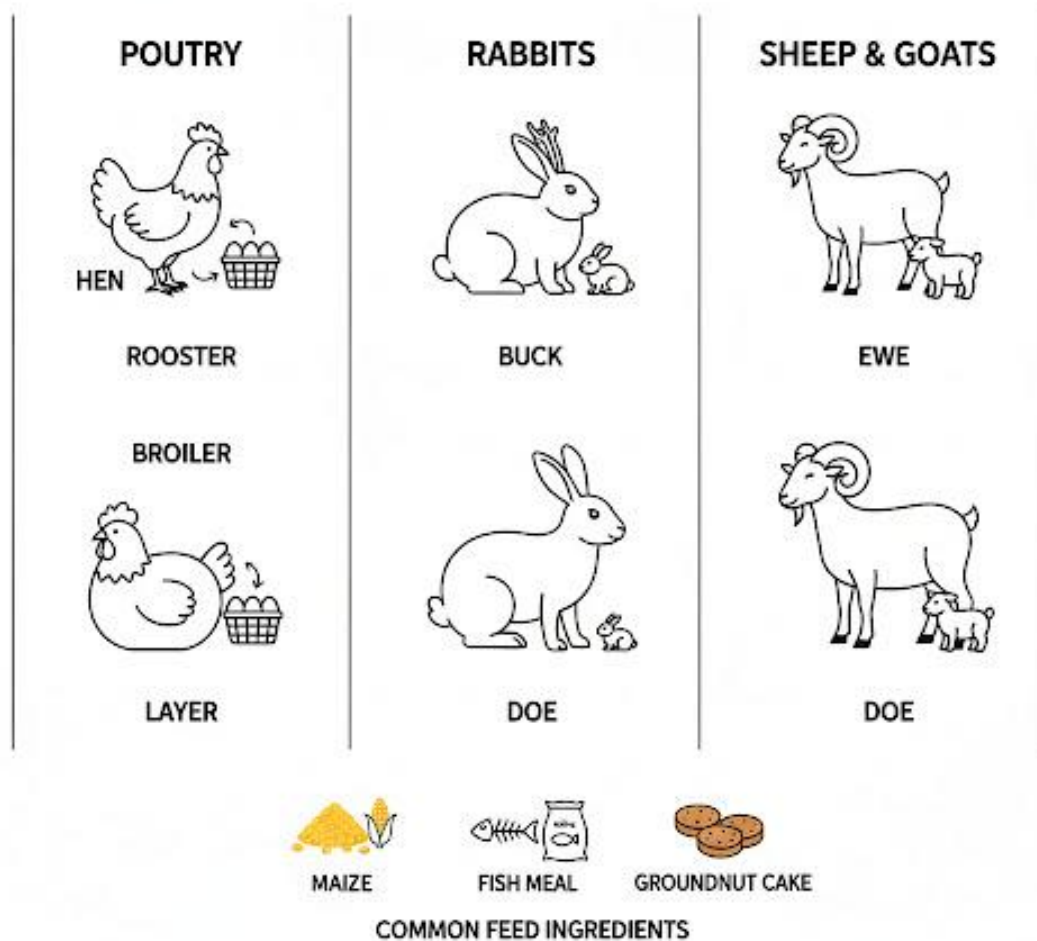
FIRST TERM

WEEK 1: REVIEW OF JSS 1 LIVESTOCK FARMING

Content/Subtopics:

- Recap of poultry, rabbit, and sheep/goat production concepts

LIVESTOCK FARMING RECAP



Performance Objectives: By the end of this lesson, students should be able to:

1. Define livestock farming and list at least five examples of farm animals.
2. Recall the key definitions and terms used in poultry, rabbitry, and sheep/goat production (e.g., hen, cock, broiler, layer, buck, doe, ewe, ram).
3. Describe the basic housing requirements (e.g., space, ventilation, sanitation) for each of these animal types.
4. List common feedstuffs used in livestock production.
5. Explain the economic importance of livestock farming in Nigeria.

Lesson Content:

1. What is Livestock Farming?

- **Definition:** Livestock farming is the branch of agriculture concerned with the raising of domestic animals (farm animals) for human use and profit.
- **Examples:** Poultry (chickens, turkeys, ducks), Rabbits (Rabbitry), Sheep (Ovine), Goats (Caprine), Cattle (Bovine), Pigs (Swine), Fish (Aquaculture), Snails (Heliciculture).

2. Recap of Key Production Concepts

- **A. Poultry Production:**
 - **Definition:** The raising of domestic birds for eggs (layers) or meat (broilers).
 - **Key Terms:**
 - **Chick:** A baby chicken.
 - **Pullet:** A young female chicken (yet to lay eggs).
 - **Cockerel:** A young male chicken.
 - **Hen:** An adult female chicken.
 - **Cock/Rooster:** An adult male chicken.
 - **Layer:** A chicken raised specifically for egg production.
 - **Broiler:** A chicken raised specifically for meat production (grows very fast).
 - **Housing:** Birds are kept in a **poultry house** or "pen." The system can be:

- **Deep Litter:** Birds are raised on the floor, which is covered with wood shavings (litter).
- **Battery Cage:** Layers are kept in individual cages for high-density egg production.
- Housing must be well-ventilated, secure from predators, and kept dry.
- **B. Rabbit Production (Rabbitry):**
 - **Definition:** The raising of rabbits for meat, fur, or as pets.
 - **Key Terms:**
 - **Buck:** An adult male rabbit.
 - **Doe:** An adult female rabbit.
 - **Kit:** A baby rabbit.
 - **Kindling:** The act of a doe giving birth.
 - **Housing:** Rabbits are kept in **hutches**. Hutches must be clean, dry, well-ventilated (but not draughty), and have a wire-mesh floor to allow droppings to fall through.
- **C. Sheep (Ovine) and Goat (Caprine) Production:**
 - **Definition:** The raising of small ruminants for meat, milk, and wool/skin.
 - **Key Terms:**
 - **Ewe:** An adult female sheep.
 - **Ram:** An adult male sheep.
 - **Lamb:** A baby sheep.
 - **Doe/Nanny:** An adult female goat.
 - **Buck/Billy:** An adult male goat.
 - **Kid:** A baby goat.
 - **Housing:** Goats and sheep are hardy. They can be raised in:
 - **Extensive System:** Allowed to roam and graze freely (common in Nigeria).
 - **Intensive System:** Kept in pens or sheds and fed by the farmer.
 - Housing must provide shelter from rain and direct sun.

3. Recap of Feeds and Feeding

- Livestock are fed **rations** (the total feed given in 24 hours). A **balanced ration** contains all essential nutrients:
 - **Carbohydrates (Energy):** From maize, guinea corn, millet, cassava.
 - **Proteins (Body-building):** From groundnut cake, soybean meal, fish meal.
 - **Fats & Oils (Energy):** From palm oil.
 - **Minerals & Vitamins (Health/Production):** From bone meal, oyster shell, vitamin premixes.
 - **Water:** Clean, fresh water is the most important nutrient.
- **Forages/Roughages:** Grasses and legumes (e.g., elephant grass, hay) are essential for ruminants (sheep, goats) and rabbits.

4. Economic Importance of Livestock

- **Food:** Provides high-quality protein (meat, eggs, milk).
- **Income:** Farmers sell animals and their products to earn money.
- **Employment:** Provides jobs for farmworkers, veterinarians, feed producers, etc.
- **Manure:** Droppings are used as organic fertilizer for crop farms.
- **Raw Materials:** Provides skin/hides for leather, wool for textiles.

Evaluation:

1. What is livestock farming? List three types of livestock farms.
2. Define the following terms: (a) Broiler, (b) Layer, (c) Buck, (d) Ewe.
3. What is the difference between a "deep litter" system and a "battery cage" system?
4. List the five essential classes of nutrients needed by farm animals.
5. State three reasons why livestock farming is important to Nigeria's economy.

Assignment:

1. Draw a simple table. In three columns, list five (5) key terms for Poultry, Rabbitry, and Sheep/Goat production.
2. What is a "balanced ration"?

3. List three examples of feed ingredients used as a source of (a) Carbohydrates and (b) Proteins.
4. In your own words, describe the ideal housing for rabbits.
5. Why is "water" considered the most important nutrient for farm animals?



WEEK 2: POULTRY MANAGEMENT (ADVANCED)

Topic: Poultry Management

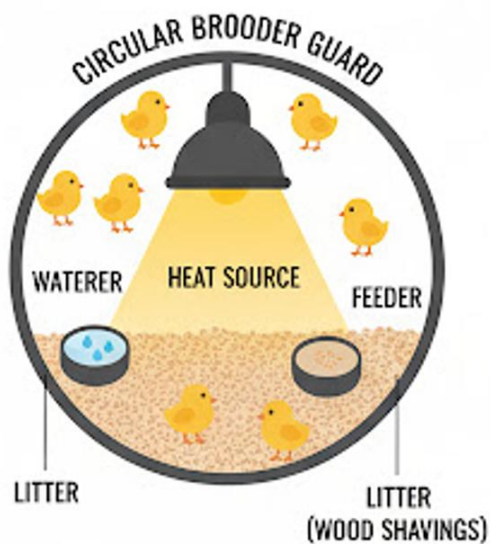
Content/Subtopics:

- Brooding management
- Culling
- Vaccination
- Record-keeping

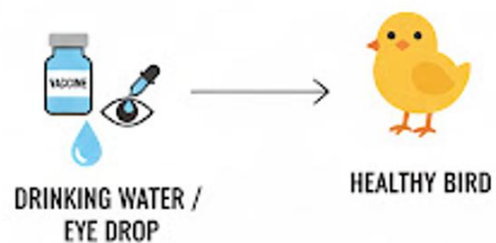
Performance Objectives: By the end of this lesson, students should be able to:

1. Define "brooding" and describe the requirements for an artificial brooder.
2. Define "culling" and list three reasons why a bird should be culled.
3. Define "vaccination" and explain its importance in disease prevention.
4. List three common poultry diseases and their vaccination times.
5. Explain the importance of record-keeping in a poultry farm.

ARTIFICIAL BROODER SETUP



VACCINATION PROTOCOL



Lesson Content:

1. Brooding Management

- **Definition:** Brooding is the care given to young birds (chicks) from one day old until they are able to regulate their own body temperature, typically from 0 to 4-6 weeks.
- **Types of Brooding:**
 - **Natural Brooding:** A mother hen provides warmth and protection for her chicks. This is only suitable for small-scale, backyard farming.
 - **Artificial Brooding:** The farmer provides the chicks' needs (heat, protection, feed, water) using equipment. This is essential for all commercial poultry.
- **Requirements for an Artificial Brooder:**
 1. **A Brooding House:** A clean, disinfected pen, free from draughts.
 2. **Litter:** The floor must be covered with dry, absorbent material like wood shavings. This provides insulation and absorbs moisture.
 3. **A "Brooder Guard":** A circular "fence" (made of cardboard or metal) that keeps the chicks close to the heat source and prevents them from straying and getting cold.
 4. **A Heat Source:** This is the most critical part.
 - *Examples:* Electric bulbs, infrared lamps, gas brooders, or charcoal stoves (used with care).
 - **Temperature:** The temperature must be 33-35°C in the first week and reduced by 2-3°C each subsequent week.
 5. **Feeders and Waterers:** Special, small-sized equipment for chicks, placed where they can easily reach them.
- **Managing the Brooder:**
 - Check the chicks' behaviour:
 - *Evenly spread:* Temperature is perfect.
 - *Huddled under the heat:* Too cold.
 - *Far away from the heat:* Too hot.

- *Huddled in one corner:* There is a cold draught.

2. Culling

- **Definition:** Culling is the process of identifying and removing unproductive, sick, or undesirable birds from the flock.
- **Importance:** It is a key management practice.
 - It stops sick birds from infecting healthy ones.
 - It saves money (you don't waste feed on birds that won't lay eggs or grow).
 - It improves the overall productivity and profitability of the farm.
- **Birds to be Culled:**
 - **Sick or Deformed Birds:** Birds showing signs of disease (drooping, coughing) or with physical deformities (crossed beak, lame).
 - **Poorly Growing Birds:** (In broilers) Birds that are significantly smaller than the rest.
 - **Unproductive Layers:** (In layers) Hens that have stopped laying eggs (identified by dry, shrunken combs/wattles and a hard, small abdomen).
 - **Old Birds:** Layers past their peak production (usually after 12-14 months of laying).
 - **Aggressive Birds:** (e.g., "egg eaters").

3. Vaccination

- **Definition:** Vaccination is the process of introducing a vaccine (a weakened or dead form of a disease-causing agent) into a bird's body to stimulate its immune system to produce antibodies. These antibodies then protect the bird from future infection by the real, live disease.
- **Importance: Prevention is better than cure.** Many poultry diseases (like Newcastle) are viral, have no cure, and can wipe out an entire flock in days. Vaccination is the only reliable way to prevent them.
- **Common Diseases and Vaccination Schedule (Example):**
 - **Marek's Disease:** Given at Day 1 (in the hatchery).

- **Newcastle Disease (ND):** Given multiple times.
 - *Intra-ocular* (eye drop) at Day 1 or Day 7.
 - *Lasota* (in drinking water) at Week 3-4.
 - Repeated boosters are given every few months.
- **Gumboro Disease (IBD):** Given in drinking water at Week 2 and Week 4.
- **Fowl Pox:** Given via *wing-web stab* at Week 6-8.
- **Methods of Vaccination:** In drinking water, eye drop, injection, or wing-web stab.

4. Record-Keeping

- **Definition:** The practice of writing down and keeping track of all important activities and data on the farm. (This will be covered in detail in Week 9).
- **Importance in Poultry:**
 - To track *feed consumption* (how much are they eating?).
 - To monitor *growth rate* (for broilers) and *egg production* (for layers).
 - To know when *vaccinations* are due.
 - To track *mortality* (deaths) and identify health problems early.
 - To calculate *profit or loss*.

Evaluation:

1. What is brooding? Differentiate between natural and artificial brooding.
2. What is the purpose of a "brooder guard"?
3. A farmer observes his chicks are all huddled directly under the heat lamp. What does this mean, and what should he do?
4. What is "culling"? List three types of birds that should be culled.
5. What is "vaccination," and why is it more important than "cure" in poultry farming?

Assignment:

1. List and explain the five (5) essential requirements for an artificial brooder.
2. Imagine you are a layer farmer. How would you identify a hen that has stopped laying eggs (an unproductive layer)?

3. What is Newcastle Disease? How is it vaccinated against?
4. Why is it important for a poultry farmer to keep records of mortality (deaths)?
5. (Research) Find the names of two other poultry diseases not mentioned in the lesson.



WEEK 3: POULTRY FEED FORMULATION

Topic: Poultry Feed Formulation

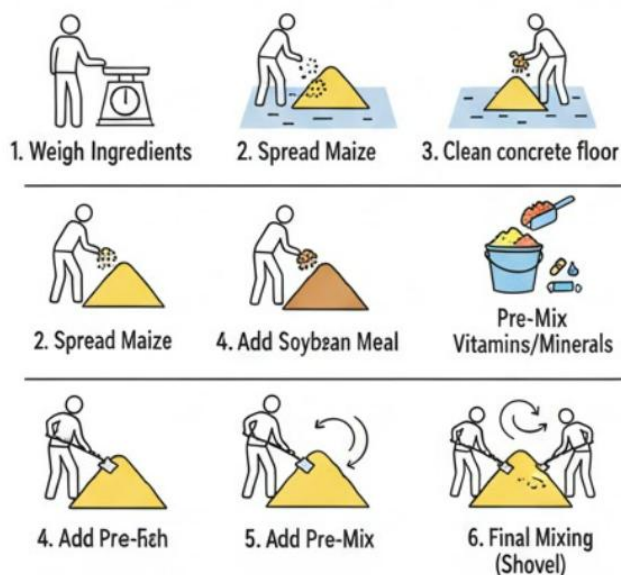
Content/Subtopics:

- Balancing nutrients
- Mixing procedures
- Feed ratios and formulation steps

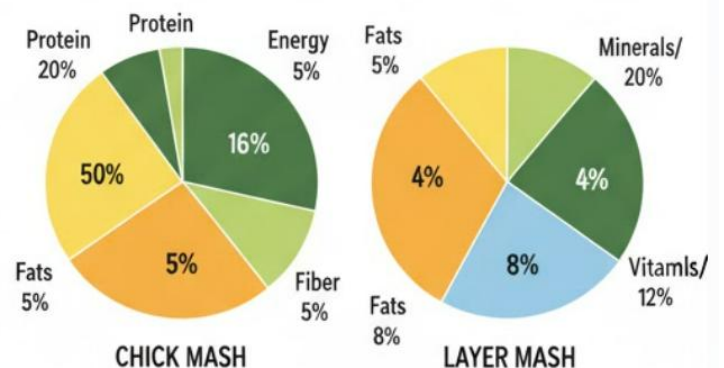
Performance Objectives: By the end of this lesson, students should be able to:

1. Define "feed formulation" and "balanced ration."
2. List the six essential classes of nutrients.
3. Identify common local ingredients for each nutrient class.
4. Explain the difference between a "starter," "grower," and "finisher/layer" mash.
5. Describe the steps for mixing feed ingredients on a farm.

POULTRY FEED MIXING PROCEDURE



NUTRIENT COMPOSITION



Lesson Content:

1. What is Feed Formulation?

- **Definition:** Feed formulation is the process of selecting, combining, and mixing different feed ingredients (like maize, soya, etc.) in specific, calculated amounts (ratios) to create a single, balanced ration.
- **Balanced Ration:** A feed that contains all six essential nutrients in the correct proportions to meet the specific needs of the animal (e.g., a chick has different needs than a laying hen).
- **Why Formulate Feed?** Commercial feeds (like "TopFeed") can be expensive. By formulating their own feed, farmers can:
 - Reduce costs.
 - Control the quality of the ingredients.
 - Utilize locally available ingredients.

2. Balancing Nutrients A balanced poultry ration must provide:

1. Carbohydrates (Energy):

- *Need:* For energy (movement, warmth, growth).
- *Ingredients:* Maize (the main source), guinea corn, millet, cassava grit.

2. Proteins (Body-building):

- *Need:* For growth, feather development, and egg production.
- *Ingredients (Plant-based):* Soybean meal, Groundnut cake (GNC).
- *Ingredients (Animal-based):* Fish meal, Blood meal.

3. Fats & Oils (Energy):

- *Need:* Concentrated source of energy.
- *Ingredients:* Palm oil, vegetable oil.

4. Minerals (Health & Structure):

- *Need:* For strong bones and eggshells.
- *Ingredients:* Bone meal (for Phosphorus), Oyster/Snail shell (for Calcium), Salt.

5. Vitamins (Health & Metabolism):

- *Need:* For disease resistance, growth, and egg production.
- *Ingredients:* Vitamin Premix (a manufactured powder containing all necessary vitamins A, B, D, E, K, etc.).

6. **Water:** (Provided separately)

3. Feed Ratios and Types of Mash The *ratio* of protein to energy changes as the bird grows.

- **A. Chick Mash (Broiler Starter / Layer Starter):**
 - *For:* Day-old chicks (0-4/6 weeks).
 - *Ratio:* **Very high protein** (approx. 20-23%) for rapid growth. Low in fibre.
- **B. Grower Mash:**
 - *For:* Pullets (6-18 weeks).
 - *Ratio:* **Medium protein** (approx. 15-18%). High in fibre. Designed to help the bird develop a strong frame, not to get fat.
- **C. Layer Mash:**
 - *For:* Laying hens (18+ weeks).
 - *Ratio:* **Medium protein** (approx. 16-18%) but **very high in Calcium** (for eggshells).
- **D. Broiler Finisher:**
 - *For:* Broilers (4/5 weeks to sale).
 - *Ratio:* **High protein** (approx. 19-21%) and **very high energy** (carbs/fats) to promote rapid weight gain (meat).

4. Mixing Procedures (On-Farm) You cannot just put all ingredients in a pile and stir. They must be mixed perfectly, or one bird will get all the protein and another will get all the maize.

- **Step 1: Gather & Weigh:** Weigh the exact amount of each ingredient (e.g., 50kg Maize, 25kg Soya, etc.).
- **Step 2: Spread Major Ingredients:** Spread the largest ingredient (Maize) on a clean, concrete floor.
- **Step 3: Add Minor Ingredients:** Spread the next largest ingredient (Soya) on top of the maize.

- **Step 4: Pre-mix Micro-ingredients:** Mix the *tiny* ingredients (premix, salt, bone meal) in a separate bucket with a few handfuls of maize. This stops them from getting lost.
- **Step 5: Add Pre-mix:** Spread the pre-mixed bucket contents over the large pile.
- **Step 6: Mix (Shoveling):** Use shovels to turn the pile over completely. Move the pile from one spot to another, then back again. Repeat this 3-4 times until the colour is uniform (no streaks of yellow maize or white premix).
- **Step 7: Bagging:** Bag the finished feed in clean, dry sacks and store off the floor.

Evaluation:

1. What is feed formulation?
2. What is a balanced ration?
3. What is the main ingredient used to supply (a) energy and (b) protein in poultry feed?
4. What is the main difference between "Chick Mash" and "Layer Mash"?
5. Why is it important to "pre-mix" tiny ingredients like vitamins?

Assignment:

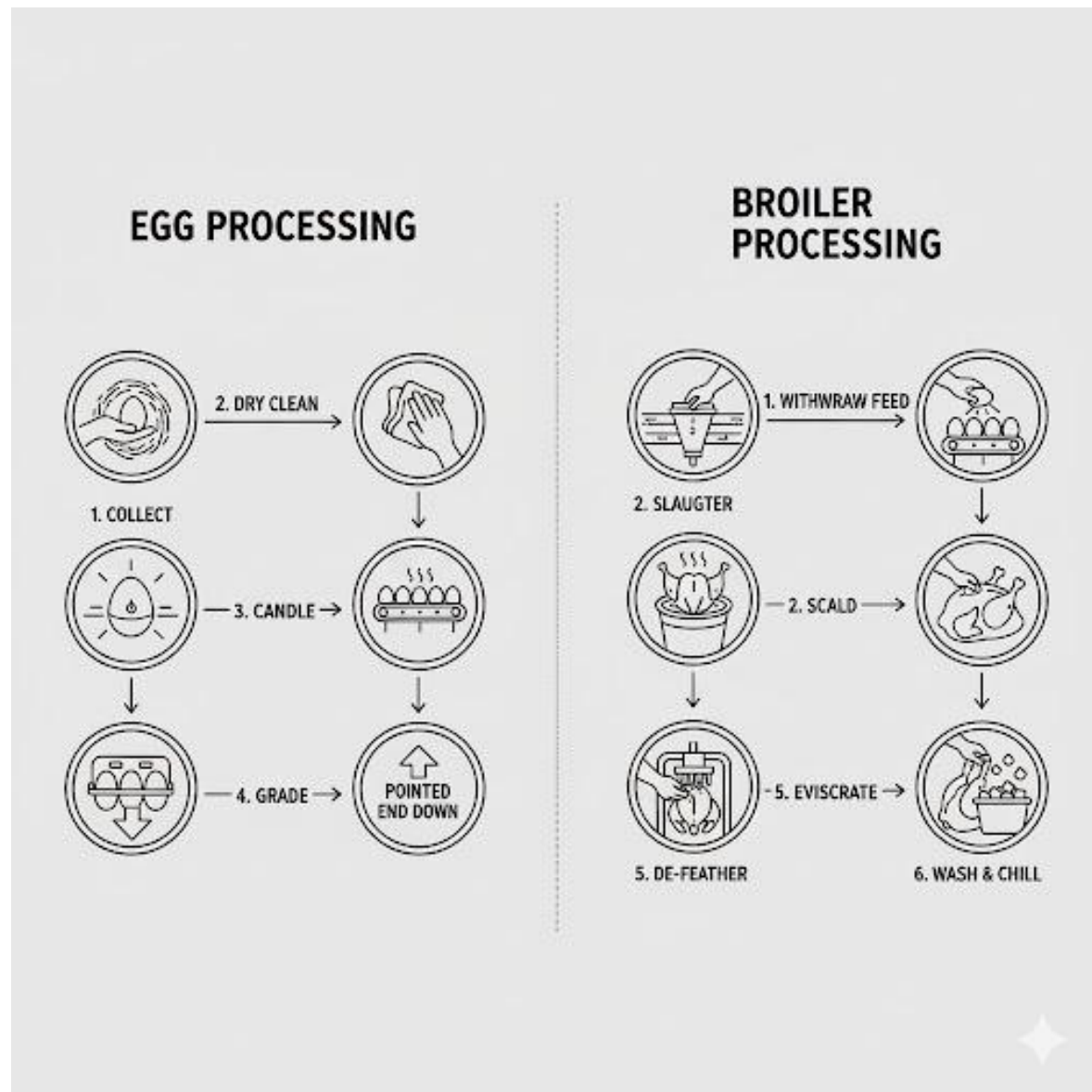
1. List the six classes of nutrients and one poultry feed ingredient for each.
2. What are the three types of feed given to a laying hen from day old to maturity?
3. Why is "Grower Mash" low in protein but "Chick Mash" is high in protein?
4. Describe the step-by-step procedure for mixing feed on a farm.
5. (Research) What is the "Pearson Square"? (Write a 2-sentence definition).

WEEK 4: EGG AND MEAT PRODUCTION

Topic: Egg and Meat Production

Content/Subtopics:

- Egg collection, processing, and storage
- Slaughtering and dressing of birds



Performance Objectives: By the end of this lesson, students should be able to:

1. Describe the correct procedure for collecting and handling eggs.
2. Explain the processes of egg candling and grading.
3. Describe the correct method for storing eggs to maintain freshness.
4. List the steps involved in slaughtering a broiler.
5. List the steps involved in "dressing" a bird (de-feathering and evisceration).

Lesson Content:

1. Egg Production and Processing (Layers)

- **A. Collection:**
 - Eggs should be collected *at least* 2-3 times per day.
 - Why? This prevents eggs from getting dirty (from manure), breaking (from other hens), or being eaten by the hens (egg-eating vice).
 - Use clean, dry egg trays or baskets.
- **B. Cleaning:**
 - Eggs should ideally be "dry-cleaned" using a clean, dry cloth or sandpaper to gently remove any dirt.
 - *Washing* eggs is discouraged because it removes the "bloom" (a natural, protective cuticle) which makes the egg spoil faster.
- **C. Candling:**
 - **Definition:** The process of shining a bright light (a "candler") through an egg to check its internal quality.
 - **It checks for:**
 - **Cracks:** Tiny hairline cracks not visible to the naked eye.
 - **Fertility:** (If there is a rooster) You can see the embryo developing.
 - **Internal Defects:** Blood spots, meat spots, or a "watery" white.
 - **Air Sac Size:** A small air sac means a fresh egg. A large air sac means an old egg.
- **D. Grading:**

- This is the process of sorting eggs by quality and size (weight).
- Eggs are graded as Grade A (Large, Medium), Grade B, etc. This helps the farmer charge different prices.
- **E. Storage:**
 - Eggs must be stored in a **cool, dry place**. (A refrigerator is ideal for home use).
 - They must be stored with the **pointed end facing down**.
 - Why? This keeps the yolk centered and prevents the large air sac (at the blunt end) from rupturing, keeping the egg fresh for longer.

2. Meat Production and Processing (Broilers)

- **A. Slaughtering (Killing):**
 - **Feed Withdrawal:** Birds are starved (no feed, only water) for 8-12 hours before slaughter. This empties the intestines and prevents contamination of the meat.
 - **Method:** The bird is hung upside down in a "killing cone." The jugular vein and carotid artery in the neck are cut with a sharp, clean knife. The bird is allowed to bleed out completely. This is the most humane and efficient method.
- **B. Dressing (Processing the Carcass):**
 - **Step 1: Scalding:** The dead bird is dipped into hot water (approx. 60°C) for 30-60 seconds. This loosens the feathers. (Do not use boiling water, as it will cook the skin).
 - **Step 2: De-feathering:** The feathers are plucked (pulled) by hand or by a "plucking machine" (a drum with rubber fingers).
 - **Step 3: Evisceration:** This is the removal of the internal organs.
 - The head, neck, and feet (shanks) are cut off.
 - A cut is made around the vent (anus).
 - The entire set of internal organs (intestines, gizzard, liver, heart) is carefully pulled out.
 - The "edible giblets" (heart, liver, gizzard) are separated, cleaned, and kept. The rest is discarded.
 - **Step 4: Washing:** The carcass (the "dressed" bird) is thoroughly washed with clean, cold water inside and out.

- **C. Storage:**

- The dressed bird must be **chilled** or **frozen** immediately to prevent bacteria from growing.
- It is then packaged in plastic bags for sale.

Evaluation:

1. Why should eggs be collected 2-3 times per day?
2. What is "candling," and what are two things it checks for?
3. How should eggs be stored (which end facing down, and why)?
4. What is "scalding" in broiler processing, and why is it done?
5. What is "evisceration"?

Assignment:

1. Describe the steps for collecting and handling fresh eggs on a farm.
2. Draw a simple diagram of an egg, showing the yolk, the air sac (at the blunt end), and the pointed end.
3. List, in order, the steps for slaughtering and dressing a broiler.
4. Why are broilers starved before slaughter?
5. (Practical/At home) Ask to see a raw egg. Identify the blunt end and the pointed end. Tell your parents the correct way to store it in the tray.

WEEK 6: RABBIT MANAGEMENT (ADVANCED)

Topic: Rabbit Management (Advanced)

Content/Subtopics:

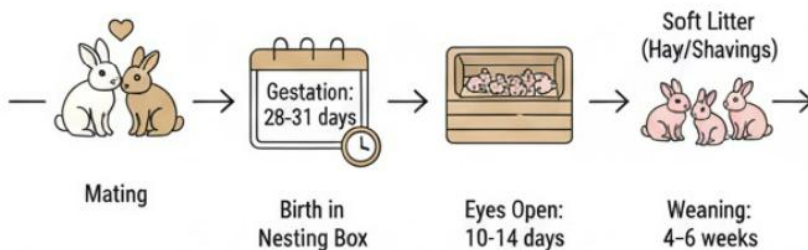
- Breeding, kindling, care of young
- Weaning and growth stages

Performance Objectives: By the end of this lesson, students should be able to:

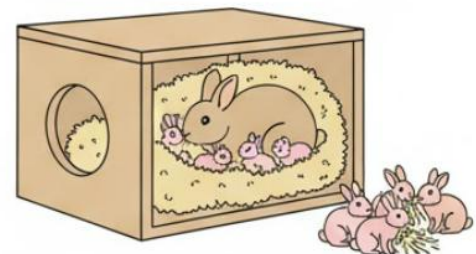
1. Define key breeding terms (buck, doe, gestation, kindling).
2. Describe the correct procedure for mating rabbits.
3. Describe the management of a doe before and during kindling (giving birth).
4. Define "weaning" and state the correct age to wean kits.
5. Describe the care of young rabbits (kits) from birth to weaning.

Lesson Content:

RABBIT BREEDING & WEANING



CORRECT NESTING BOX SETUP



1. Breeding Management

- **Key Terms:**
 - **Buck:** Adult male rabbit (for breeding).

- **Doe:** Adult female rabbit (for breeding).
- **Gestation Period:** The length of pregnancy in a rabbit (approx. **28-31 days**).
- **Mating Process:**
 1. **Check for Heat:** A doe does not have a regular "heat" cycle, but she has periods of receptiveness. A doe "in heat" will have a swollen, reddish-purple vulva.
 2. **Take the Doe to the Buck:** **ALWAYS** take the female (Doe) to the male's (Buck's) hutch.
 3. *Why?* A doe is very territorial and will fight the buck if he is brought to her hutch. The buck's hutch is his territory, and he will mate quickly.
 4. **Observe Mating:** Mating is very fast (a few seconds). The buck will mount the doe and will often fall over on his side with a grunt when mating is complete.
 5. **Return Doe:** Immediately remove the doe and return her to her hutch.
 6. **Record Keeping:** Write down the date of mating.

2. Kindling Management

- **Definition: Kindling** is the act of a doe giving birth to her litter of kits.
- **The Nesting Box:**
 - About 2-3 days *before* the due date (around Day 28), the farmer must provide a **nesting box**.
 - This is a small, quiet box (wooden or metal) placed inside the doe's hutch.
 - The farmer should half-fill it with clean, soft, dry litter (wood shavings or hay).
- **Doe's Behaviour:** The doe will use the box to build a nest. She will arrange the hay and then *pull fur from her own chest* to create a soft, warm lining for the kits. This is a normal and important behaviour.
- **During Kindling:**
 - The doe must be left alone in a quiet, stress-free environment.
 - She will usually give birth at night.
 - Litter size can be 4-10 kits.
 - Kits are born **altricial** (blind, deaf, and furless).

3. Care of Young (Kits) from Birth to Weaning

- **Day 1 (Post-Kindling):**
 - Calmly check the nesting box.
 - Remove any dead or deformed kits.
 - Count the live kits and record the number.
 - Check their bellies: A full, rounded belly means the doe is feeding them. A wrinkled, empty belly is a sign of trouble.
- **Day 1-14 (Eyes Closed):**
 - Check the kits *briefly* once a day.
 - Ensure the nest is clean and dry.
 - Do not disturb the doe or the kits unnecessarily.
 - The kits' eyes will open around Day 10-14.
- **Day 14-28 (Exploring):**
 - The kits will start to come out of the nesting box and explore the hutch.
 - They will begin to nibble on their mother's feed and hay.
 - The nesting box can be removed around Day 21 (3 weeks) as it will be dirty.

4. Weaning and Growth Stages

- **Definition: Weaning** is the process of separating the kits from their mother (the doe) so they can live and grow independently.
- **When to Wean:** This is usually done at **4 to 6 weeks** of age.
 - *Early weaning (4 weeks)* is stressful but allows the doe to be re-bred faster.
 - *Later weaning (6 weeks)* is less stressful for the kits and produces stronger animals.
- **How to Wean:**
 - The best method is to **remove the doe** from the hutch and leave the kits in the hutch they are familiar with. This is less stressful.
 - The weaned kits (now called "weaners") are then grouped by size and sex (if possible) into "growing hutches."
- **Managing Weaners:**

- This is a high-stress period. Weaners must be given high-quality feed (pellets or forages) and *constant* access to clean water.
- They are raised as "growers" until they reach market weight (around 3-4 months).

Evaluation:

1. Define "Kindling" and "Gestation Period."
2. Why must you always take the doe to the buck's hutch for mating, and not the other way around?
3. What is a nesting box, and when should you provide it?
4. How does a doe prepare her nest for the kits?
5. What is "weaning," and at what age is it typically done?

Assignment:

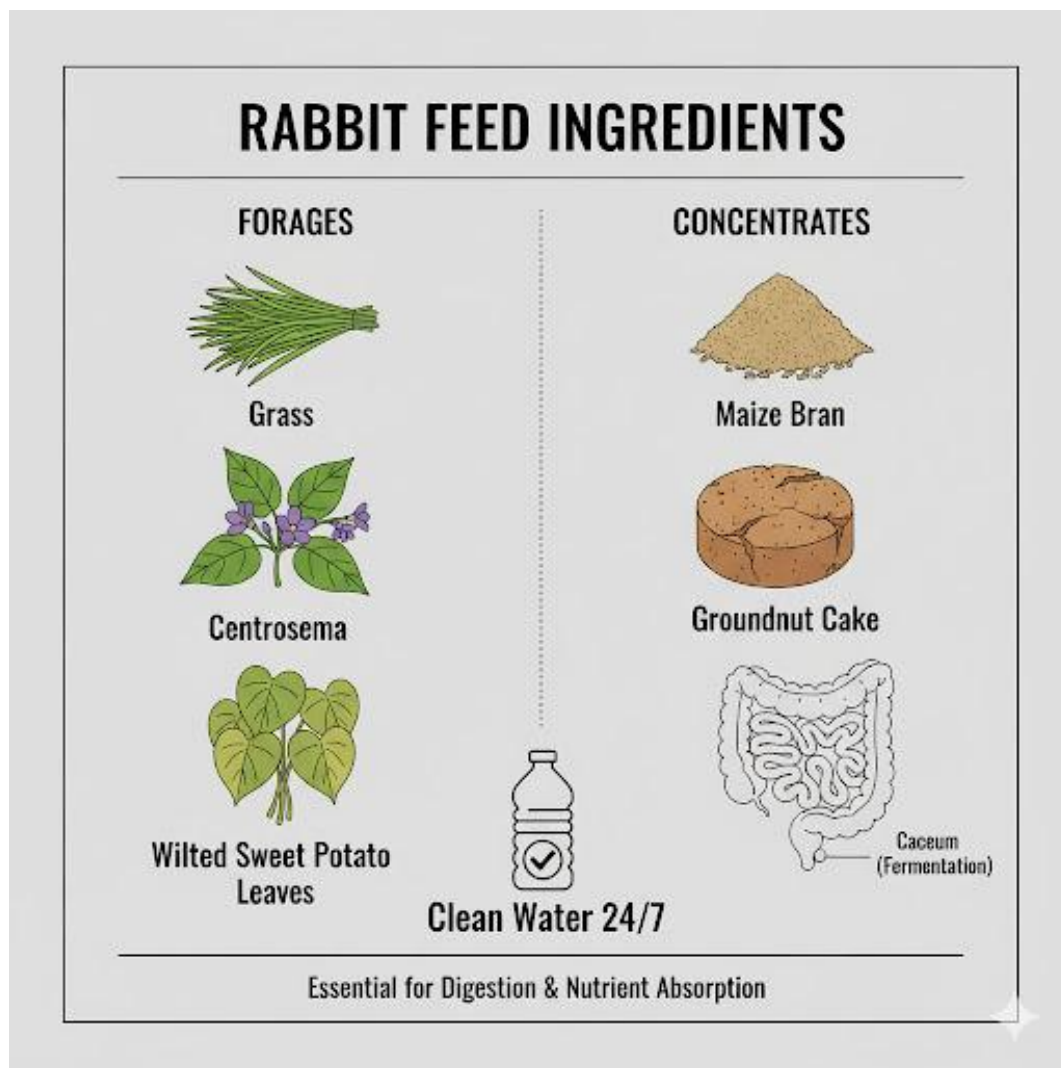
1. List and define the following terms: Buck, Doe, Kit, Kindling, Weaning.
2. Describe the step-by-step process for mating rabbits.
3. What is the gestation period of a rabbit?
4. Describe the daily checks a farmer should perform on a litter of kits for the first week.
5. What is the best method for weaning, and why is it considered the least stressful for the kits?

WEEK 8: RABBIT FEED FORMULATION

Topic: Rabbit Feed Formulation

Content/Subtopics:

- Local feed ingredients
- Feeding management and schedules



Performance Objectives: By the end of this lesson, students should be able to:

1. Identify rabbits as "herbivores."
2. List three examples of forages (grasses/legumes) for feeding rabbits.

3. List three examples of concentrate ingredients for feeding rabbits.
4. Describe the advantages and disadvantages of using local feed ingredients.
5. Outline a proper feeding schedule for a rabbit.

Lesson Content:

1. Understanding the Rabbit's Digestive System

- Rabbits are **Herbivores**, meaning they are designed to eat plants.
- They are also "**Hind-gut Fermenters**." This means they have a special organ called the **caecum** that acts like a fermentation vat to break down tough, fibrous plant material (like grass).
- **Caecotrophy**: Rabbits produce two types of droppings: hard, round pellets (waste) and soft, "night faeces" called **caecotropes**. They *eat* these caecotropes directly from their anus to absorb the nutrients (B-vitamins and proteins) that were produced in the caecum. This is normal and essential for their health.

2. Local Feed Ingredients for Rabbits Rabbit feed must be high in **fibre** and balanced with protein and energy.

- **A. Forages (Roughages) - The Most Important Part (Provide Fibre):**
 - *Grasses*: Elephant grass, Guinea grass, Giant star grass, Maize stems.
 - *Legumes (High in protein)*: Centrosema, Calopogonium (Calopo), Stylo, Gliricidia (These are often grown as "cover crops").
 - *Leaves*: Banana leaves, pawpaw leaves, sweet potato leaves.
 - **Rule**: Forages should be *wilted* (left in the sun for a few hours) before feeding. This reduces moisture and prevents bloating/diarrhoea.
- **B. Concentrates (Provide Energy and Protein):**
 - *Energy*: Maize bran, rice bran, wheat offal, cassava peels (must be sun-dried).
 - *Protein*: Groundnut cake (GNC), Soybean meal, Palm kernel cake (PKC).
- **C. Commercial Feed:**

- **Pellets:** These are commercially formulated, balanced rations. They are easy to feed and ensure good growth, but they are expensive. Many farmers use a *combination* of pellets and forages.

3. Advantages and Disadvantages of Local Ingredients

- **Advantages:**

- They are very cheap (forages are often free).
- They are readily available in the local environment.
- They provide essential fibre for the rabbit's digestion.

- **Disadvantages:**

- They can be bulky and low in energy (rabbits must eat a lot).
- They are not always "balanced" (may lack certain nutrients).
- Risk of poisoning: Some plants are toxic (e.g., wild cassava, some ornamental flowers).
- They can introduce diseases or pesticides if not collected from a safe source.

4. Feeding Management and Schedules

- **Water: Clean, fresh water must be available 24/7.** A rabbit will die from lack of water faster than from lack of food.
- **Feeding Schedule (Example):**
 - **Morning (7-9 a.m.):**
 1. Clean all feeders and waterers.
 2. Provide fresh water.
 3. Provide the daily ration of *concentrates* or *pellets*. (e.g., 50-100g per rabbit, depending on size).
 - **Afternoon (1-3 p.m.):**
 1. Provide a large handful of *wilted forages* (grasses/legumes).
 2. Check water again.
 - **Evening (5-6 p.m.):**
 1. Remove any uneaten forages from the afternoon.

2. Provide *another* large handful of fresh forages (rabbits are most active at night).
- **Rule: NEVER** give mouldy or rotten feed. **NEVER** feed lawnmower clippings (they ferment and kill rabbits).

Evaluation:

1. What does it mean that a rabbit is a "herbivore"?
2. What are "forages"? Give two examples of forages you can feed a rabbit.
3. What are "concentrates"? Give two examples.
4. Why should forages be "wilted" before feeding?
5. What is the *most* important rule for a rabbit's feeding schedule? (Clean water 24/7).

Assignment:

1. Define "Herbivore" and "Caecotrophy."
2. List three (3) advantages and two (2) disadvantages of using local forages for rabbits.
3. Outline a suitable daily feeding schedule for an adult rabbit.
4. Why should a farmer *not* feed lawnmower clippings to a rabbit?
5. (Practical) Collect, identify, and bring to class (in a bag) 3 different types of safe forages for rabbits from your school or home environment.

WEEK 9: RECORD KEEPING IN LIVESTOCK PRODUCTION

Topic: Record Keeping in Livestock Production

Content/Subtopics:

- Types and importance of records
- Feeding, health, and financial records

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "record keeping."
2. State five key reasons why record-keeping is important for a farm.
3. List and describe four different types of farm records.
4. Differentiate between a "production record" and a "health record."
5. Analyze a simple record to make a management decision.

ESSENTIAL FARM RECORDS

**PRODUCTION RECORD**

Date	Bird Count	Eggs Collected

**HEALTH RECORD**

Date	Vaccine / Drug Given

**FEEDING RECORD**

Date	Feed Type	Quantity Used (kg)

**FINANCIAL RECORD**

INCOME	EXPENSES
Sales	Feed
	Medication
Other	Other
TOTAL INCOME	
PROFIT = INCOME - EXPENSES	

Simple, Organized, Essential for Success.

Lesson Content:

1. What is Record Keeping?

- **Definition:** Record keeping is the systematic, regular, and accurate documentation (writing down) of all activities and data related to the farm business.
- **Analogy:** A farm record book is like a student's "Report Card" or a ship's "Logbook." It tells you where you have been, where you are, and where you are going.
- **Rule:** A farm without records is a *hobby*, not a *business*. You are just "guessing."

2. Importance of Record Keeping

1. **For Decision Making (The main reason):** Records provide *facts* (data) so you can make smart decisions.
 - *Example:* Records show Hen A lays 6 eggs/week and Hen B lays 2 eggs/week.
Decision: Cull Hen B.
2. **To Track Profit and Loss:** Records show how much you spent (expenses) and how much you earned (income). This is the only way to know if your farm is profitable.
3. **To Monitor Health:** A sudden drop in egg production or feed intake (seen in your records) is the *first sign* of disease, before you even see a sick bird.
4. **For Breeding:** Records show which buck/doe produce the most/strongest kits, or which ram/ewe produces fast-growing lambs. You use this to select your *best* animals for breeding.
5. **To Secure Loans:** If you want to get a loan from a bank to expand your farm, you *must* show them your records (financial, production) to prove your business is serious and profitable.

3. Types of Farm Records

- **A. Production Records (What the animals are *producing*):**

- *Poultry (Layers)*: Daily egg collection (total eggs, % production), number of cracked eggs.
- *Poultry (Broilers)*: Weekly body weight, growth rate.
- *Rabbits*: Mating dates, kindling dates, number of kits born (alive/dead), weaning weights.
- *Sheep/Goats*: Birth dates, birth weights, weaning weights.
- **B. Health Records (Disease and Treatment):**
 - Vaccination dates and type of vaccine.
 - Dates of any treatment (what drug was given, for what sickness).
 - Mortality (Death) Record: Date of death, animal number, suspected cause.
- **C. Feeding Records (What you are *spending*):**
 - Type of feed bought.
 - Quantity of feed bought.
 - Cost of feed.
 - Daily feed consumption (how much is the flock eating?).
- **D. Financial Records (The *Business* side):**
 - **Expenses**: A list of everything you *spend* money on (Cost of chicks/kits, feed, drugs, salaries, transport, new equipment).
 - **Income (Revenue)**: A list of everything you *sell* (Crates of eggs, live broilers, culled layers, rabbits, manure).
 - **Profit/Loss Statement**: A summary sheet (e.g., at the end of the month).
 - **Profit = Total Income - Total Expenses**

4. Example of a Simple Record Card (Rabbitry): | Doe ID: 001 | Hutch No: A5_ | Breed: New Zealand White | | :--- | :--- | :--- | | **Mating Date** | **Buck ID** | **Kindling Date** | **# Born Alive** | **# Born Dead** | **Weaning Date** | **# Weaned** | | Jan 10 | B-02 | Feb 11 | 8 | 1 | Mar 25 | 7 | | Apr 1 | B-01 | May 2 | 7 | 0 | Jun 13 | 7 |

- **Analysis:** This doe (001) is a good mother. She has large litters (8, 7) and weans almost all of them. We should keep her.

Evaluation:

1. What is record keeping?
2. State two reasons why keeping records is essential for a farm business.
3. What is the difference between a "Production Record" and a "Health Record"?
4. What information would you find in a "Financial Record"?
5. What is the formula for calculating Profit?

Assignment:

1. Why is a farm without records considered a "hobby" and not a "business"?
2. List the four (4) main types of farm records discussed in the lesson.
3. Design a simple "Daily Egg Production Record" for a poultry farmer. (It should have columns for: Date, Number of Birds, Eggs Collected (Morning), Eggs Collected (Evening), Total Eggs, Remarks).
4. What is a "Mortality Record," and why is it an important part of the Health Record?
5. If a farmer's Total Income for a month is N50,000 and his Total Expenses are N35,000, what is his profit or loss?



WEEK 10: SANITATION AND BIOSECURITY

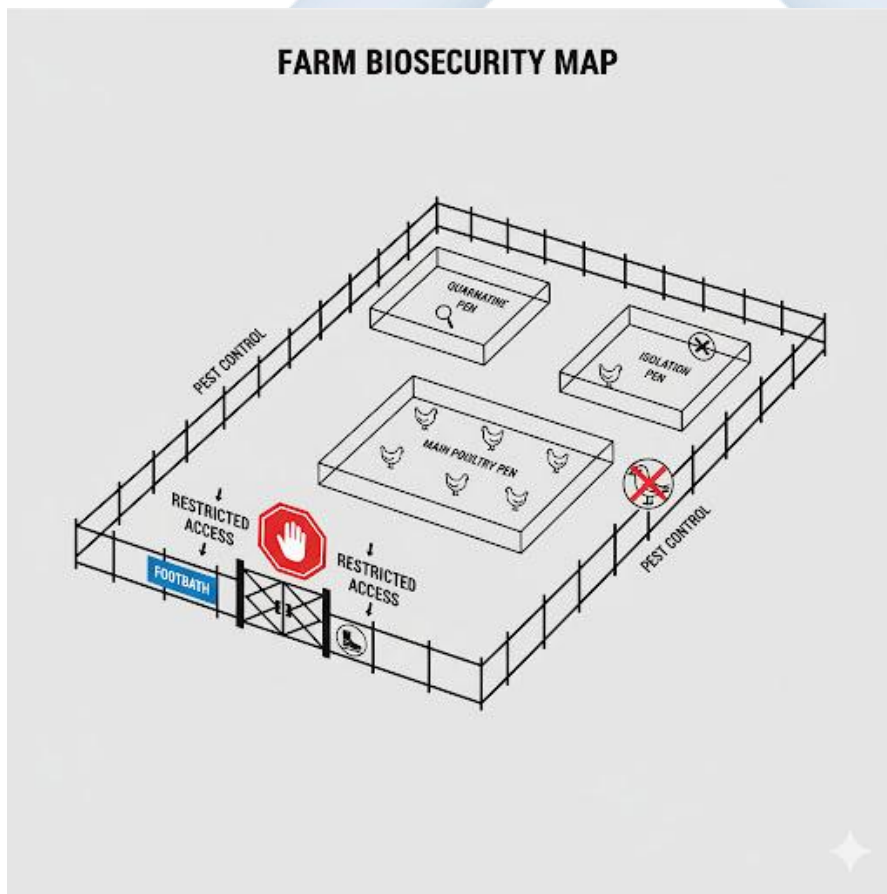
Topic: Sanitation and Biosecurity

Content/Subtopics:

- Hygiene, disinfection, and disease prevention methods

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Sanitation" and "Hygiene."
2. Define "Biosecurity" and state its main goal.
3. List three key hygiene practices on a farm.
4. List and describe three main biosecurity measures.
5. Explain the purpose of a "footbath" (foot dip).



Lesson Content:

1. Definitions

- **Hygiene:** Refers to the *practices* and *conditions* that help to maintain health and prevent the spread of disease. It is the overall "cleanliness" of the farm.
- **Sanitation:** Refers to the *specific actions* of cleaning and *disinfecting* to remove or kill germs (pathogens).
 - *Example:* **Hygiene** is the *policy* of keeping the farm clean. **Sanitation** is the *action* of sweeping, washing, and spraying disinfectant.
- **Biosecurity:**
 - **Definition:** A set of preventive measures designed to reduce the risk of infectious diseases being *introduced* into, or *spreading* within, a farm.
 - **Goal:** To build a "security fence" around the farm to keep germs *OUT* and stop them from spreading *IN*.

2. Hygiene and Sanitation Practices Good hygiene is a daily job.

1. Daily Cleaning:

- Feeders and waterers must be washed *daily*. Stale, wet feed grows mould. Dirty water spreads disease.
- Remove spilled feed.

2. Manure Management:

- Remove manure (droppings) regularly. In a deep litter, this means "turning" the litter. In a cage/hutch, this means scraping the droppings from under the cage.
- A buildup of manure produces ammonia (a sharp smell) which damages birds' lungs.

3. Disinfection:

- **Definition:** The process of using chemicals (disinfectants) to kill viruses and bacteria.

- **"All-In, All-Out" System:** The *best* sanitation. A farmer raises *one* batch of birds, sells *all* of them, then completely cleans and *disinfects* the empty pen before the *new* batch arrives. This breaks the disease cycle.

4. Pest Control:

- Rats, mice, and wild birds are major carriers of disease (and steal feed).
- The farm must have traps, bait stations, and netting to keep them out.

3. Biosecurity Methods (Disease Prevention) This is about controlling *movement*.

• 1. Isolation and Quarantine:

- **Isolation:** If a bird *inside* your farm gets sick, you must *isolate* it (move it to a "sick bay" pen) immediately to stop it from infecting others.
- **Quarantine:** When you buy *new* animals, you must keep them in a separate "quarantine pen" for at least 2-3 weeks to watch them for signs of sickness *before* they are allowed to join your main flock.

• 2. Restricting Access (People and Vehicles):

- The farm should be fenced.
- Limit visitors. People (like feed truck drivers or other farmers) can carry diseases on their shoes and clothing.
- **Footbath:** A shallow tray at the entrance of the farm/pen, filled with disinfectant. *Everyone* must dip their boots in it to kill germs before entering.

• 3. Flock Management:

- Keep birds of different *ages* separate.
- Do not mix chickens with turkeys or ducks (they can share diseases).
- Properly dispose of dead birds (mortality) by burying or burning them far from the pen. Do not eat or sell sick birds.

• 4. Vaccination: (As covered in Week 2) This is a key part of biosecurity.

Evaluation:

1. What is the difference between "hygiene" and "sanitation"?
2. What is "biosecurity," and what is its main goal?

3. Why must feeders and waterers be cleaned daily?
4. What is the "All-in, All-out" system, and why is it a good practice?
5. What is a "footbath," and what is its purpose?

Assignment:

1. Define Hygiene, Disinfection, and Biosecurity.
2. What is the difference between "Isolation" and "Quarantine"?
3. Why are rats and wild birds a threat to a poultry farm?
4. Describe three biosecurity measures you would put in place if you started your own farm.
5. A farmer buys 10 new hens. What is the *first* thing he should do with them? (Quarantine them).



WEEK 11: REVISION

Topic: Revision

Content/Subtopics:

- Review of advanced livestock management topics

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key terms from the term.
2. Explain the processes of brooding, culling, and vaccination.
3. Describe the key steps in feed formulation and mixing.
4. Compare and contrast egg/meat processing.
5. Explain the advanced management of rabbits (breeding, kindling, weaning).
6. Discuss the importance of record-keeping and biosecurity.

Lesson Content:

1. Summary of Term 1 Topics (Advanced Management):

- **Week 1 (Review):** We remembered the basics of poultry, rabbits, sheep, and goats.
- **Week 2 (Poultry Management):**
 - **Brooding:** Caring for day-old chicks with an artificial heat source (35°C).
 - **Culling:** Removing sick, old, or unproductive birds.
 - **Vaccination:** Preventing diseases (like Newcastle, Gumboro) by giving a vaccine.
- **Week 3 (Poultry Feed):**
 - **Formulation:** Mixing ingredients (Maize, Soya, Premix) in the right ratios.
 - **Ratios:** Chick Starter (High Protein), Grower (Med Protein), Layer (High Calcium), Broiler (High Energy).
- **Week 4 (Production):**
 - **Eggs:** Collect, candle, grade, and store (pointed end down).
 - **Meat:** Slaughter (bleed), Scald (hot water), De-feather, Eviscerate (remove organs), Chill.

- **Week 6 (Rabbit Management):**
 - **Breeding:** Take Doe to Buck. Gestation is 31 days.
 - **Kindling:** Doe gives birth in a nesting box.
 - **Weaning:** Separate kits from doe at 4-6 weeks.
- **Week 8 (Rabbit Feed):**
 - Rabbits are **Herbivores**. They *must* have fibre (forages like grass).
 - Feed must include forages, concentrates, and *constant* clean water.
- **Week 9 (Record Keeping):**
 - The "brain" of the farm. If it's not written down, it didn't happen.
 - Types: Production, Health, Feeding, Financial.
 - Used to track profit and make decisions.
- **Week 10 (Biosecurity):**
 - Keeping the farm clean (sanitation) and safe from disease.
 - Methods: Footbaths, quarantine (for new animals), isolation (for sick animals).

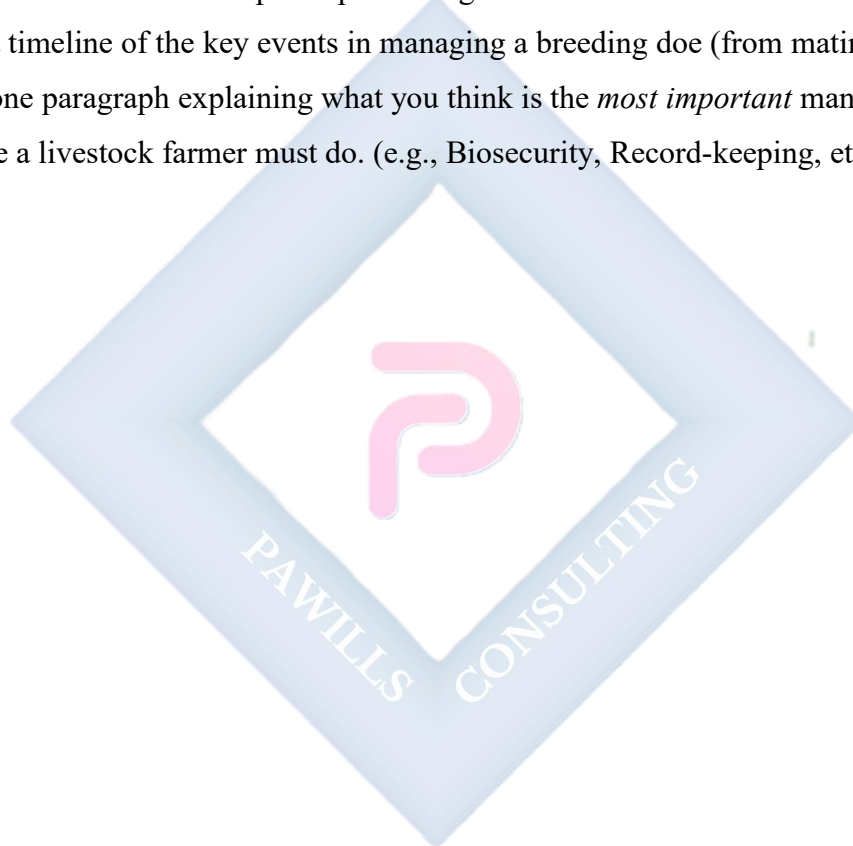
2. Class Activity: Final Q&A (Teacher asks rapid-fire questions from the term to prepare for the final examination).

- "What is brooding?"
- "Why do we cull?"
- "Why do we vaccinate?"
- "What is the main energy ingredient in feed?"
- "What is the main protein ingredient?"
- "What is candling?"
- "What is evisceration?"
- "What is the gestation period of a rabbit?"
- "What is kindling?"
- "What is weaning?"
- "What is the most important nutrient for a rabbit?"
- "What are the 4 types of farm records?"
- "What is a footbath for?"
- "What is the difference between quarantine and isolation?"

Evaluation: This week's evaluation is the First Term (Final) Examination.

Assignment:

1. Study all notes from the term.
2. Make a list of 10 key terms from this term and their definitions (e.g., Brooding, Culling, Vaccination, Formulation, Candling, Kindling, Weaning, Record-Keeping, Biosecurity, Quarantine).
3. Draw a flowchart of the steps for processing a broiler for meat.
4. Draw a timeline of the key events in managing a breeding doe (from mating to weaning).
5. Write one paragraph explaining what you think is the *most important* management practice a livestock farmer must do. (e.g., Biosecurity, Record-keeping, etc.) and why.



SECOND TERM - JSS 2 LIVESTOCK FARMING LESSON NOTES

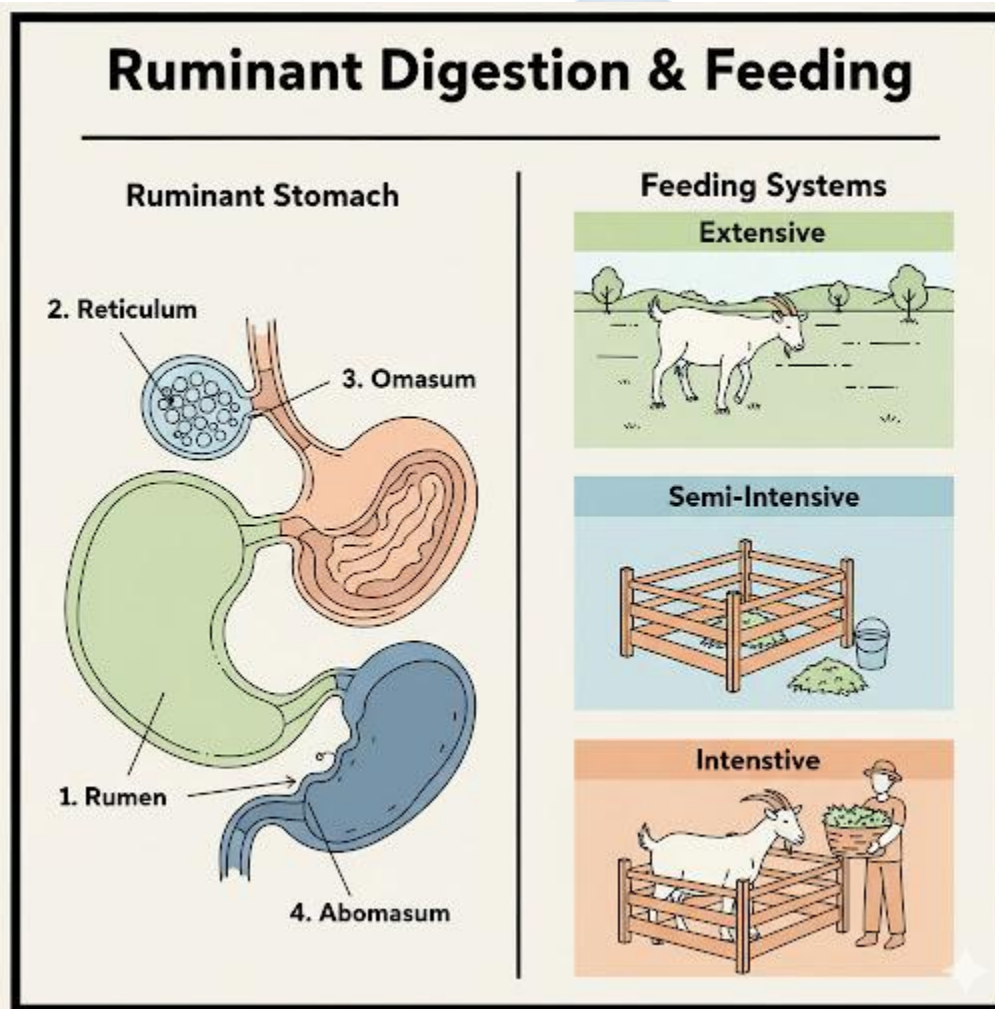


WEEK 1: SHEEP AND GOAT PRODUCTION (ADVANCED)

Topic: Sheep and Goat Production (Advanced)

Content/Subtopics:

- Feeding methods
- Breeding systems
- Care of lambs and kids



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "ruminant" and briefly describe their digestive system.
2. Differentiate between feeding methods: extensive, semi-intensive, and intensive.

3. Identify signs of heat (estrus) in ewes and does.
4. Describe the correct management of newborn lambs and kids, including the importance of colostrum.
5. Define common management practices like disbudding, castration, and creep feeding.

Lesson Content:

1. Understanding Sheep (Ovine) and Goats (Caprine) as Ruminants

- **Definition:** Sheep and goats are **ruminants**. This means they have a complex, four-chambered stomach (Rumen, Reticulum, Omasum, Abomasum).
- **Function:** The **Rumen** is the largest chamber and acts as a large fermentation vat. It contains billions of microbes (bacteria and protozoa) that can break down tough, fibrous plant materials (like grass and hay) that non-ruminants (like humans or pigs) cannot digest.
- **Implication for Feeding:** This digestive system means their primary feed *must* be high in fibre (roughage/forage).

2. Feeding Methods for Sheep and Goats

- **A. Extensive System (Traditional Grazing/Pastoralism):**
 - **Method:** Animals are allowed to roam or "graze" freely over large, unfenced areas (pastures) to find their own food (grasses, shrubs, leaves). This is the most common system in Nigeria.
 - **Advantages:** Very low cost (feed is free), low labour requirement.
 - **Disadvantages:** High risk of theft, exposure to diseases and predators, poor control over nutrition, animals can be lost.
- **B. Semi-Intensive System:**
 - **Method:** A combination. Animals are allowed to graze during the day, but they are brought back to a pen or shed in the evening.
 - In the pen, they are given **supplementary feed** (like groundnut cake, maize bran, or salt licks) and clean water.
 - **Advantages:** Balances low cost with better security and nutrition.

- **Disadvantages:** Requires more labour and capital for housing and extra feed.
- **C. Intensive System (Zero Grazing / Cut-and-Carry):**
 - **Method:** Animals are permanently confined in a pen or house.
 - The farmer *cuts* forages (grasses, legumes) and *carries* them to the animals.
 - A full, balanced ration (including concentrates) is provided.
 - **Advantages:** Full control over nutrition, high security from theft/predators, easy to collect manure.
 - **Disadvantages:** Very high cost (feed and housing), very high labour requirement (cutting grass daily), higher risk of disease spread within the pen.

3. Breeding Systems and Management

- **Goal:** To produce strong, healthy lambs and kids.
- **Key Terms:**
 - **Estrus ("Heat"):** The period when the female (ewe or doe) is receptive to the male (ram or buck) and will stand to be mated.
 - **Gestation Period (Pregnancy):** Approximately **145-150 days** (about 5 months) for both sheep and goats.
- **Signs of Heat:**
 - Restlessness, continuous bleating.
 - Shaking or "flagging" of the tail.
 - Swollen, reddish vulva.
 - The female stands still to be mounted by the male.
- **Mating:** Can be **Natural Mating** (letting the ram/buck run with the females) or **Controlled Mating** (bringing a specific female to a specific male, to record the date).

4. Care of Newborn Lambs and Kids This is the most critical period. High mortality can occur here.

1. At Birth:

- **Clear Airways:** Ensure the mucus is cleared from the newborn's nose and mouth.

- **Navel Cord Care:** The umbilical cord should be cut (if necessary) and immediately disinfected by dipping it in **Iodine solution** or methylated spirit. This prevents **Navel Ill**, a deadly infection.
2. **Colostrum (First Milk):**
- The *most important* step. The newborn *must* drink the mother's first milk (colostrum) within the first 6 hours of life.
 - **Why?** Colostrum is not just food; it is a "natural vaccine" rich in antibodies that gives the newborn its first immunity against diseases.
3. **Identification:** The animal should be tagged (ear tag) or notched for record-keeping.
4. **Creep Feeding:**
- This is the practice of providing a special, high-protein "starter" feed for the young animals in an area (a "creep feeder") that the mothers *cannot* access. This helps the young grow faster and reduces stress at weaning.
5. **Disbudding and Castration:**
- **Disbudding:** Removing the small horn "buds" from kids (goats) when they are 1-2 weeks old, using a hot iron. This prevents the growth of dangerous horns.
 - **Castration:** Removing the testicles of male animals that are *not* intended for breeding. This makes them grow fatter (meat quality) and reduces aggressive behaviour.

Evaluation:

1. What is a ruminant?
2. What is the main difference between the "extensive" and "intensive" feeding systems?
3. What is "estrus," and what is one sign of it in a goat?
4. What is colostrum, and why is it essential for a newborn lamb?
5. Define "disbudding" and "creep feeding."

Assignment:

1. Draw a simple diagram of the four-chambered stomach of a ruminant.
2. List two advantages and two disadvantages of the "Semi-Intensive" system.

3. What is the approximate gestation period for a sheep or goat?
4. Describe the step-by-step care a farmer should give to a kid or lamb in the first hour after it is born.
5. (Research) Find the name of one breed of sheep and one breed of goat commonly found in Nigeria.



WEEK 2: DISEASE PREVENTION AND CONTROL

Topic: Disease Prevention and Control

Content/Subtopics:

- Vaccination schedules
- Sanitation
- Isolation of sick animals

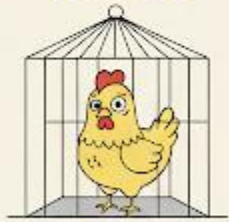
Livestock Vaccination & Bioscourity

Livestock Vaccination Schedule

Day 1				Week 2			Week 6	
								
Day 1: Marek's				Week 2: Gumbotro			Week 6: Fowl Pox	

Biosecurity: Isolation vs Quarantine

Isolation



Sick animal, kept alone.

Quarantine



New animals, kept separate (2-4 weeks)

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "disease" and differentiate between infectious and non-infectious diseases.
2. Explain the importance of a vaccination schedule.

3. Define "sanitation" and list three key sanitation practices.
4. Differentiate between "isolation" and "quarantine."
5. Describe other control methods like deworming and dipping.

Lesson Content:

1. What is Disease?

- **Definition:** A disease is any condition that impairs (stops) the normal functioning of an animal's body.
- **Types:**
 - **Infectious Diseases:** Caused by pathogens (germs) like viruses, bacteria, and fungi. They can *spread* from one animal to another (e.g., Newcastle, PPR).
 - **Non-Infectious Diseases:** Cannot be spread. They are caused by:
 - *Nutritional:* Lack of a nutrient (e.g., rickets from lack of calcium).
 - *Metabolic:* A problem with the body's processes (e.g., bloat).
 - *Physical:* An injury (e.g., a broken leg).
- **Golden Rule: Prevention is always better (and cheaper) than cure.**

2. Vaccination Schedules

- **Definition:** A vaccination schedule is a *timetable* or *calendar* that tells the farmer *which* vaccine to give, *what age* to give it, and *how* to give it (e.g., injection, in water).
- **Importance:** It is a planned program to build immunity in the entire flock or herd *before* a disease outbreak occurs.
- **Key Livestock Diseases and Vaccines:**
 - **PPR (Peste des Petits Ruminants):**
 - *Affects:* Sheep and goats (highly fatal "goat plague").
 - *Vaccine:* PPR vaccine, given to lambs/kids at 3-4 months old.
 - **Fowl Pox:**
 - *Affects:* Poultry.
 - *Vaccine:* Fowl Pox vaccine (wing-web stab), given at 6-8 weeks.
 - **Newcastle Disease (ND):**

- *Affects:* Poultry (highly fatal).
- *Vaccine:* Lasota (in water) or Intra-ocular (eye drop), given multiple times.
- **Gumboro (IBD):**
 - *Affects:* Poultry (chicks).
 - *Vaccine:* Gumboro vaccine (in water), given at 2 and 4 weeks.

3. Sanitation (Review)

- **Definition:** Sanitation refers to the specific actions of *cleaning* and *disinfecting* to remove dirt and kill germs.
- **Key Practices:**
 1. **Daily Cleaning:** Washing all feeders and waterers with soap and water before refilling.
 2. **Regular Manure Removal:** Scraping and removing droppings from pens (daily or weekly) to keep the environment dry and reduce ammonia.
 3. **Disinfection:** Using chemical disinfectants (like Izal, JIK, or specialized farm chemicals) to wash the pen, especially after an outbreak or between batches of animals ("all-in, all-out").
 4. **Footbaths:** (A biosecurity measure) Placing a tray of disinfectant at the entrance of the farm for all visitors and workers to dip their boots in.

4. Isolation and Quarantine (Biosecurity) This is about controlling the *movement* of animals to control disease.

- **Isolation (for *sick* animals):**
 - **Definition:** The separation of animals that are *already sick* from the rest of the healthy flock.
 - **Action:** Immediately move any animal showing signs of sickness (e.g., drooping, coughing, diarrhoea) to a separate "sick bay" or "isolation pen" for treatment. This stops them from infecting the others.
- **Quarantine (for *new* animals):**

- **Definition:** The separation of *newly purchased* animals (that *look healthy*) from the main flock.
- **Action:** When you buy a new goat or new chickens, they must be kept in a "quarantine pen" far away from your main animals for at least 2-3 weeks.
- **Purpose:** To observe them for any signs of disease that they might be carrying from their old farm.

5. Other Control Methods

- **A. Parasite Control (Internal):**
 - **Deworming (Drenching):** The regular giving of anti-parasite medicine (anthelmintics) to animals (especially ruminants) to kill internal worms (like roundworms, tapeworms).
- **B. Parasite Control (External):**
 - **Dipping or Spraying:** The use of insecticides (acaricides) to kill external parasites like *ticks*, *lice*, and *mites* on the skin of cattle, sheep, and goats.

Evaluation:

1. What is the difference between an infectious and a non-infectious disease?
2. What is a vaccination schedule?
3. What is PPR? Which animals does it affect?
4. What is the main difference between "Isolation" and "Quarantine"?
5. What is "deworming" (drenching) used to control?

Assignment:

1. Define "Disease," "Sanitation," and "Biosecurity."
2. Why is "prevention" better than "cure" in livestock farming?
3. You buy 5 new goats. What is the *first* and *most important* thing you should do with them when you get to your farm?
4. You notice one of your chickens is coughing and has its eyes closed. What should you do *immediately*?

5. What is the purpose of a "footbath" at a farm entrance?



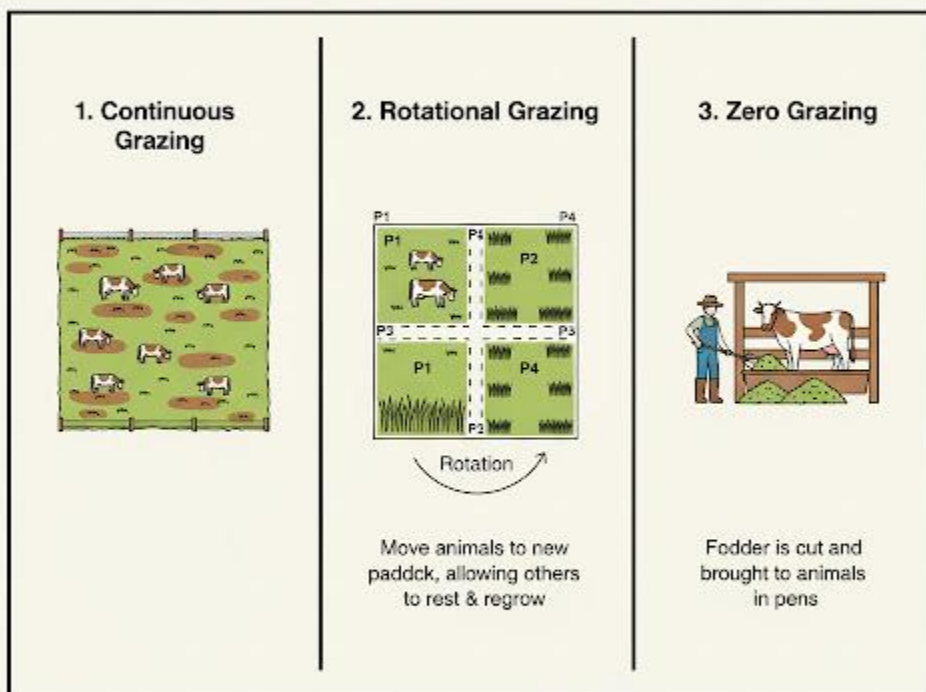
WEEK 3: PASTURE MANAGEMENT

Topic: Pasture Management

Content/Subtopics:

- Types of pastures
- Grazing systems
- Establishment and maintenance of pastures

Grazing Systems



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "pasture" and "forage."
2. Differentiate between natural and artificial (planted) pastures.

3. List three common pasture grasses and two common pasture legumes in Nigeria.
4. Describe and compare continuous, rotational, and zero grazing systems.
5. Explain the key steps in establishing and maintaining a pasture.

Lesson Content:

1. What is Pasture?

- **Definition:** A pasture is a large area of land covered with grasses, legumes, and other plants (called **forages**), which are grown for animals (especially ruminants like cattle, sheep, and goats) to graze on.
- **Forage:** The plant material (leaves and stems) available to be eaten by the animals.

2. Types of Pastures

- **A. Natural Pasture (Native Pasture):**
 - **Definition:** These are grasslands that grow naturally, without being planted by humans. They contain a mix of different "wild" grasses, legumes, and weeds.
 - **Location:** This is the most common pasture in Nigeria (e.g., savannah grasslands).
 - **Advantages:** No cost of establishment.
 - **Disadvantages:** Low quality and low quantity of feed, especially in the dry season. The grasses are often tough and not very nutritious.
- **B. Artificial Pasture (Planted/Cultivated Pasture):**
 - **Definition:** This is a pasture that is *planted* by a farmer. The farmer chooses *specific*, high-quality, nutritious grasses and legumes to plant.
 - **Advantages:** High yield (produces more feed), high nutritional value (better for animals), feed is available for longer.
 - **Disadvantages:** Very high cost and labour for establishment (clearing, ploughing, planting).

3. Common Pasture Plants (Forages)

- **Grasses (Mainly for Energy):**

- Elephant Grass (*Pennisetum purpureum*)
- Guinea Grass (*Panicum maximum*)
- Giant Star Grass (*Cynodon nlemfuensis*)
- **Legumes (Mainly for Protein - they "fix" nitrogen):**
 - *Centrosema* (Centro)
 - *Stylosanthes* (Stylo)
 - *Calopogonium* (Calopo)
- **Ideal Pasture:** A good artificial pasture is a *mix* of grasses and legumes.

4. Grazing Systems This is the *management* of how animals use the pasture.

- **A. Continuous Grazing:**
 - **Method:** Animals are kept in one large pasture area and are allowed to graze there all year round.
 - **Problem:** Leads to **overgrazing**. The animals eat their favourite grasses until they die, allowing weeds to take over.
- **B. Rotational Grazing (Paddock Grazing):**
 - **Method:** The pasture is divided into smaller sections called **paddocks**, using fences.
 - The animals are "rotated": they graze in Paddock 1 for a week, then are moved to Paddock 2.
 - This allows Paddock 1 to rest and *re-grow* (a "rest period").
 - **Advantages:** This is the *most efficient* system. It prevents overgrazing and allows the pasture to produce more feed.
- **C. Zero Grazing (Cut-and-Carry):**
 - **Method:** (As seen in Week 1). The animals are kept in a pen. The farmer cuts the grass from the pasture and brings it to them.
 - **Advantages:** No energy lost by animals walking, no trampling of grass, manure is easily collected.

5. Establishment and Maintenance of Pastures

- **Establishment (How to plant):**

1. **Site Selection:** Choose suitable, well-drained, fertile land.
2. **Land Preparation:** Clear bushes, plough, and harrow the land to create a good seedbed (just like for a crop farm).
3. **Planting:** Plant the grass/legume seeds or stems at the beginning of the rainy season.

- **Maintenance (How to care for it):**

1. **Weed Control:** Remove unwanted weeds that compete with the grasses.
2. **Fertilizer Application:** Apply NPK fertilizer (for grasses) or Superphosphate (for legumes) to boost growth.
3. **Pest and Disease Control:** Control pests like armyworms.
4. **Controlled Grazing:** Implement a rotational grazing system to prevent overgrazing.

Evaluation:

1. What is the difference between a "natural pasture" and an "artificial pasture"?
2. List two common pasture grasses and one common pasture legume.
3. What is "rotational grazing," and why is it better than "continuous grazing"?
4. What is "overgrazing"?
5. What are the first two steps in *establishing* a new pasture?

Assignment:

1. Define "Pasture" and "Forage."
2. List three advantages of an artificial (planted) pasture.
3. Draw a simple diagram showing a field divided for "rotational grazing" (at least 4 paddocks).
4. What is the "zero grazing" system?
5. List three (3) activities a farmer must do to *maintain* a pasture.

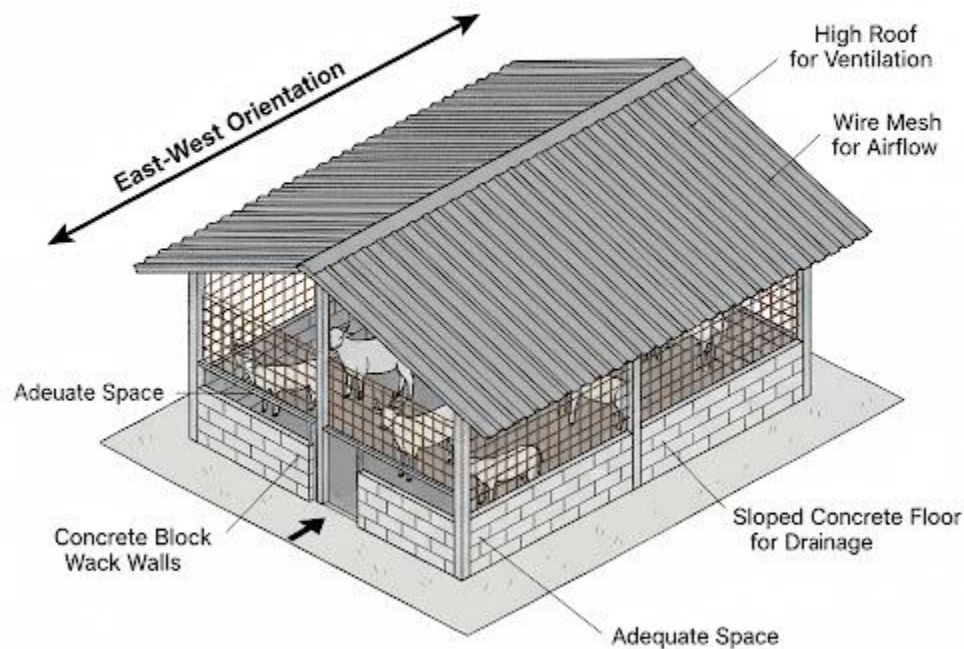
WEEK 4: LIVESTOCK HOUSING

Topic: Livestock Housing

Content/Subtopics:

- Improved housing materials and layout for animal comfort

Well-Designed Livestock Pen



Performance Objectives: By the end of this lesson, students should be able to:

1. State three reasons why good housing is important for livestock.
2. List five key *features* of a good livestock house (e.g., ventilation, drainage).
3. Identify common local materials and improved materials used for construction.
4. Define "siting" and "orientation" in farm layout.

5. Draw a simple layout of a small livestock farm.

Lesson Content:

1. Importance of Livestock Housing Housing (or a "pen") is a major factor in successful livestock farming.

1. **Protection from Weather:** It shields animals from extreme sun, heavy rain, and cold winds, which cause stress and sickness.
2. **Protection from Predators and Theft:** It provides security against dangers like snakes, dogs, foxes, and human thieves.
3. **Control and Management:** It keeps animals in one place, making it easy for the farmer to feed, water, observe, treat, and control breeding.
4. **Disease Control:** It prevents animals from mixing with wild or sick animals.

2. Features of a Good Livestock House A good house must be comfortable for the animals.

1. **Good Ventilation:** Air must be able to flow *in* and *out* (cross-ventilation) to remove hot, stale air and ammonia. This is done with open sides covered in wire mesh.
2. **Adequate Space:** Must not be overcrowded. Overcrowding causes stress, fighting, and rapid spread of disease.
3. **Good Drainage:** The floor must be *sloped* slightly, and made of materials (like concrete) that can be easily washed. There should be gutters to carry waste water away from the pen.
4. **Lighting:** Must be well-lit (natural or artificial) so the farmer can inspect the animals and the animals can see their feed.
5. **Sanitation:** Must be easy to clean. Walls and floors should be smooth, with no cracks where ticks or germs can hide.
6. **Durability:** Must be built with strong materials that can last long and withstand animal activity.

3. Housing Materials

- **A. Local/Traditional Materials:**
 - **Floor:** Rammed earth (cheap, but hard to clean, can get damp).
 - **Walls:** Wood, bamboo, mud blocks.
 - **Roofing:** Thatch (cheap, keeps the house cool, but must be replaced often and can hide pests).
- **B. Improved/Modern Materials:**
 - **Floor: Concrete** (best). It is durable, easy to wash, easy to disinfect, and stops rodents from burrowing.
 - **Walls:** Concrete blocks (dwarf walls), **wire mesh** or "chain-link" fencing (for ventilation).
 - **Roofing: Corrugated iron or aluminium sheets** (durable, last long, but can be hot if the roof is not high).

4. Farm Layout: Siting and Orientation

- **Siting (Location):**
 - Where to *locate* the farm.
 - It must be on a **well-drained** piece of land (not a swampy area).
 - It should be accessible (near a road) but *far away* from residential areas (due to smell and flies).
 - It must have a reliable source of clean water.
- **Orientation (Direction):**
 - How to *position* the building.
 - In a hot country like Nigeria, livestock houses should be built with the *long axis* in an **East-West direction**.
 - **Why?** This means the long walls (with the ventilation) face North and South. The small walls (gable ends) face East and West. This ensures that the hot, direct morning and afternoon sun *never* shines directly into the pen, keeping the animals cool.

5. Simple Farm Layout A good farm is organized for biosecurity and efficiency.

- [Simple Diagram: Fence, Gate, Footbath, Office/Store (near gate), Farm House, Poultry Pens, Rabbit Hutches, Isolation Pen (far corner), Manure Pit (downwind corner)]
- The **Isolation Pen** must be *far away* from the main pens.
- The **Feed Store** must be secure and rodent-proof.

Evaluation:

1. List three (3) reasons why livestock need good housing.
2. List four (4) features of a good livestock house.
3. What is the *best* material to use for the floor of a modern poultry pen, and why?
4. What is "orientation"?
5. In Nigeria, why should a poultry house be built in an East-West direction?

Assignment:

1. Define "Siting" and "Orientation" in farm layout.
2. List three (3) "improved" building materials and three (3) "local" materials for building a pen.
3. What is the main advantage of a concrete floor over an earth floor?
4. Why is good ventilation so important in a livestock house?
5. Draw a simple, labelled diagram of a small farm layout, including a fence, a gate, a footbath, a feed store, and an isolation pen.

WEEK 6: CONCEPT OF ENTREPRENEURSHIP

Topic: Concept of Entrepreneurship

Content/Subtopics:

- Meaning of entrepreneurship
- Characteristics of an entrepreneur



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "entrepreneur" and "entrepreneurship."

2. Differentiate between an "entrepreneur" and a "subsistence farmer."
3. Explain the role of "risk-taking" in entrepreneurship.
4. List and describe at least five key characteristics of a successful entrepreneur.
5. Identify an entrepreneur in their local community.

Lesson Content:

1. Meaning of Entrepreneur and Entrepreneurship

- **Entrepreneur:** An **entrepreneur** is a *person* who identifies a **need, problem, or opportunity** in society, organizes the resources (capital, labour, land) to create a **business (solution)**, and takes a personal and financial **risk** to launch and manage that business, with the primary goal of making a **profit**.
- **Entrepreneurship:** This is the *process* or *act* of being an entrepreneur. It is the journey of planning, starting, and running a new business.

2. Entrepreneur vs. Subsistence Farmer

- **Subsistence Farmer:** A farmer who grows crops or raises animals *just to feed their own family*. They have little or no surplus (extra) to sell.
- **Livestock Entrepreneur (Agri-preneur):** A farmer who raises animals as a **business**. Their goal is to produce a surplus (e.g., 5,000 broilers) to sell to the public for a profit. They are market-focused.

3. Key Characteristics of an Entrepreneur What makes an entrepreneur different?

1. **Risk-Taker:** This is the most important trait. They are willing to risk their *own money* and *time* on an idea that might fail. (e.g., spending N500,000 to start a poultry farm, knowing that disease could wipe it out).
2. **Innovative and Creative:** They find new or better ways of doing things.
 - *Example:* A regular farmer just sells live chickens. An *innovative* entrepreneur processes, freezes, and packages the chicken, or even opens a restaurant that sells the chicken.

3. **Hardworking and Persistent:** They work long hours and do not give up easily when they face challenges (like disease, high feed costs, etc.).
4. **Good Manager (Organized):** They must be able to manage money (budgeting), time (meeting deadlines), and people (staff).
5. **Proactive and Opportunity-Seeking:** They don't wait for opportunities; they *look* for them.
 - *Example:* They see that a new housing estate is being built and think, "Those new residents will need eggs and meat. I will start a farm nearby."
6. **Independent (Self-Reliant):** They have a desire to be their own boss and take responsibility for their own success or failure.

Evaluation:

1. What is an "entrepreneur"?
2. What is "entrepreneurship"?
3. What is the main difference between a subsistence farmer and a livestock entrepreneur?
4. What is the *most important* characteristic of an entrepreneur? (Risk-taker).
5. List three other characteristics of an entrepreneur.

Assignment:

1. Define "Entrepreneur" and "Entrepreneurship" in your own words.
2. What does it mean to be a "risk-bearer"?
3. Why is "creativity" important for an entrepreneur?
4. Name one person in your community (or in Nigeria) who you think is a successful entrepreneur (in any field, e.g., Dangote, or a local business owner).
5. List two characteristics that make that person an entrepreneur.

WEEK 8: IMPORTANCE OF ENTREPRENEURSHIP

Topic: Importance of Entrepreneurship

Content/Subtopics:

- Role in employment creation
- Opportunities in livestock farming

Performance Objectives: By the end of this lesson, students should be able to:

1. Explain how entrepreneurship leads to employment creation.
2. Explain how entrepreneurship contributes to national development (GDP).
3. Define "self-reliance" and its importance.
4. List at least five specific business opportunities in the livestock farming value chain.
5. Differentiate between production, processing, and support (service) opportunities.



Lesson Content:

1. Importance of Entrepreneurship (to the Nation)

- **A. Employment Creation:**
 - This is the most significant impact. When an entrepreneur starts a business (e.g., a poultry farm), they cannot do all the work alone.
 - They hire *farm attendants* (to feed/clean), a *farm manager*, a *veterinarian* (to vaccinate), a *driver* (to deliver eggs), and *sales staff* (to work in the shop).
 - This reduces the level of unemployment in the country.
- **B. National Development (Grows the Economy):**
 - **Food Security:** Livestock entrepreneurs produce the meat, milk, and eggs that feed the nation, reducing the need to import food.
 - **Contribution to GDP:** The profit from the business and the salaries paid to workers are all part of the nation's Gross Domestic Product (GDP).
 - **Pays Taxes:** Successful businesses pay taxes to the government, which the government then uses to build roads, schools, and hospitals.
- **C. Promotes Self-Reliance:**
 - Entrepreneurship empowers individuals to create their own "job" instead of waiting for the government or a big company to hire them. This leads to personal financial independence.
- **D. Fosters Innovation:**
 - Entrepreneurs are always looking for new, cheaper, and better ways to do things, which makes the whole industry more efficient.

2. Business Opportunities in the Livestock Farming Value Chain The "value chain" is every step from "farm to table." You can start a business at any link.

- **A. Production (The Farm):**
 - **Broiler Production:** (Raising chickens for meat).
 - **Layer Production:** (Producing eggs for sale).
 - **Rabbitry:** (Raising rabbits for meat).

- **Goat/Sheep Fattening:** (Buying thin animals, feeding them well, and selling them fat for festivals).
- **Catfish Farming:** (Aquaculture).
- **Snail Farming:** (Heliciculture).
- **B. Processing (Adding Value):**
 - **Meat Processing:** Setting up a small "slaughter and dress" shop.
 - **Frozen Food Shop:** Buying, processing, and selling frozen chicken, fish, etc.
 - **Egg Distribution:** Buying eggs in bulk from farms and selling them in crates to shops.
 - **Food Services:** Starting a restaurant or "local food" joint that *uses* the farm products.
- **C. Support / Services (Input Supply):**
 - **Feed Formulation & Sales:** Starting a "feed mill" or a shop that sells feed.
 - **Equipment Fabrication:** Welding and building cages, hutches, feeders, and waterers.
 - **Veterinary Services:** Becoming a veterinarian or opening a "vet shop" to sell drugs and vaccines.
 - **Consultancy:** Being an experienced expert who other farmers pay for advice.

Evaluation:

1. What is the most important role of entrepreneurship in a country's economy? (Employment creation).
2. How does entrepreneurship promote "self-reliance"?
3. What is the "livestock value chain"?
4. List two business opportunities in *production*.
5. List two business opportunities in *processing* or *support services*.

Assignment:

1. Define "food security" and "GDP."

2. In your own words, explain how one entrepreneur starting a 1,000-bird poultry farm can create jobs for at least 3 other people.
3. Look at the list of business opportunities. Which one would you be most interested in starting, and why?
4. What is the difference between a "production" business and a "processing" business?
5. (Research) Find out the name of one large livestock (poultry) company in Nigeria (e.g., Obasanjo Farms, Zartech, etc.).

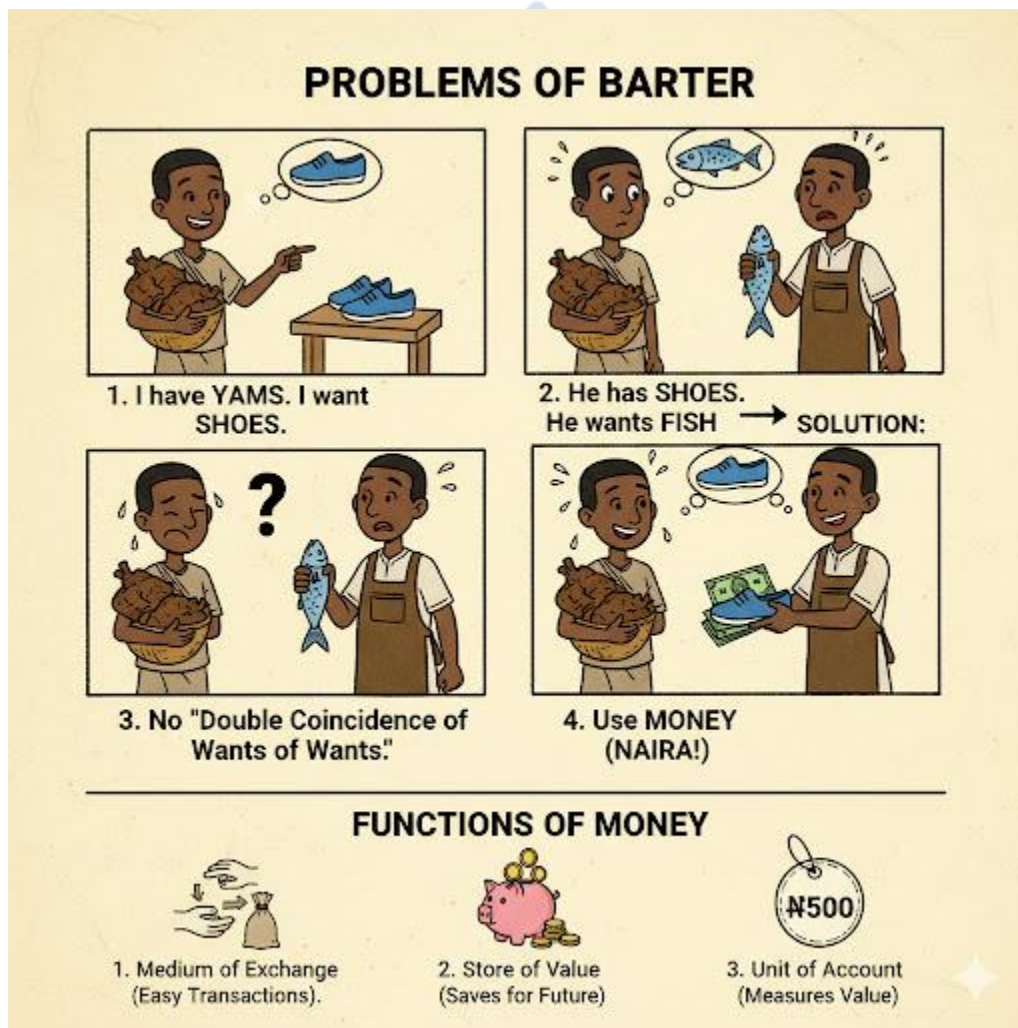


WEEK 9: CONCEPT OF MONEY

Topic: Concept of Money

Content/Subtopics:

- Definition and functions of money
- Forms of money in agribusiness.



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "money" and "barter."
2. Explain the problems associated with the barter system.
3. List and explain the four main functions of money.
4. Identify different forms of money used in modern agribusiness.

Lesson Content:

1. What is Money?

- **Definition:** Money is any *item* or *verifiable record* that is **generally accepted** as payment for goods and services or as repayment of debts in a particular country or socio-economic context.
- **Before Money: Trade by Barter**
 - **Definition:** Barter is the direct *exchange* of goods for other goods, without the use of money.
 - *Example:* "I will give you five yams if you give me one goat."
- **Problems with Barter:**
 - **Double Coincidence of Wants:** For a trade to happen, you must find someone who *has* what you want *and* *wants* what you have. (The goat owner might not want yams; he might want a shoe).
 - **No Standard Measure of Value:** How many yams is one goat worth? How many chickens is one goat worth? It's confusing.
 - **Indivisibility of Goods:** You can't "divide" a live goat if you only want two yams.
 - **Hard to Store Value:** Yams and goats will spoil or die.

2. Functions of Money (Money Solves These Problems) Money is "good" because it performs four key functions:

1. Medium of Exchange:

- This is its main job. Money is the "go-between." The goat farmer sells his goat for *money*, and then uses the *money* to buy yams. He doesn't need to find a yam farmer who wants a goat.

2. Unit of Account (Measure of Value):

- Money provides a standard "price tag." We know that a goat is N15,000 and a yam is N1,000. It makes it easy to compare the value of different items.

3. **Store of Value:**

- You can *save* money. It doesn't rot or spoil like yams (though *inflation* can reduce its value). You can sell your goat today and spend the money next year.

4. **Standard of Deferred Payment:**

- This means money can be used for *credit and debt*. You can take a loan (borrow money) today and agree to pay it back (plus interest) in six months. This is very hard to do with goats.

3. **Forms of Money in Agribusiness (Livestock Farming)** How a farmer pays for things:

• **A. Cash:**

- Physical notes (Naira) and coins. Used for everyday small transactions (e.g., paying for transport, buying drugs).

• **B. Bank Deposits (Cheques and Transfers):**

- Money held in a bank account.
- Used for large, safe transactions (e.g., paying for 500 bags of feed, paying a salary).
- **Cheque:** A written order to the bank to pay a specific amount.
- **Bank Transfer (Mobile/USSD):** Using a phone app to send money directly from one account to another. This is now the most common.

• **C. Credit (Trade Credit):**

- A form of "deferred payment." The feed seller agrees to give the farmer feed *now*, and the farmer agrees to *pay later* (e.g., after selling the broilers).

Evaluation:

1. What is "trade by barter"?
2. What is the "double coincidence of wants"?
3. What is the main function of money? (Medium of exchange).
4. How does money act as a "store of value"?

5. List two forms of money (besides cash) that a farmer can use.

Assignment:

1. Define money in your own words.
2. List three problems associated with the barter system.
3. List and *explain* the four functions of money.
4. What is "trade credit"?
5. Why is a cheque or bank transfer safer for a farmer than paying N500,000 in cash?

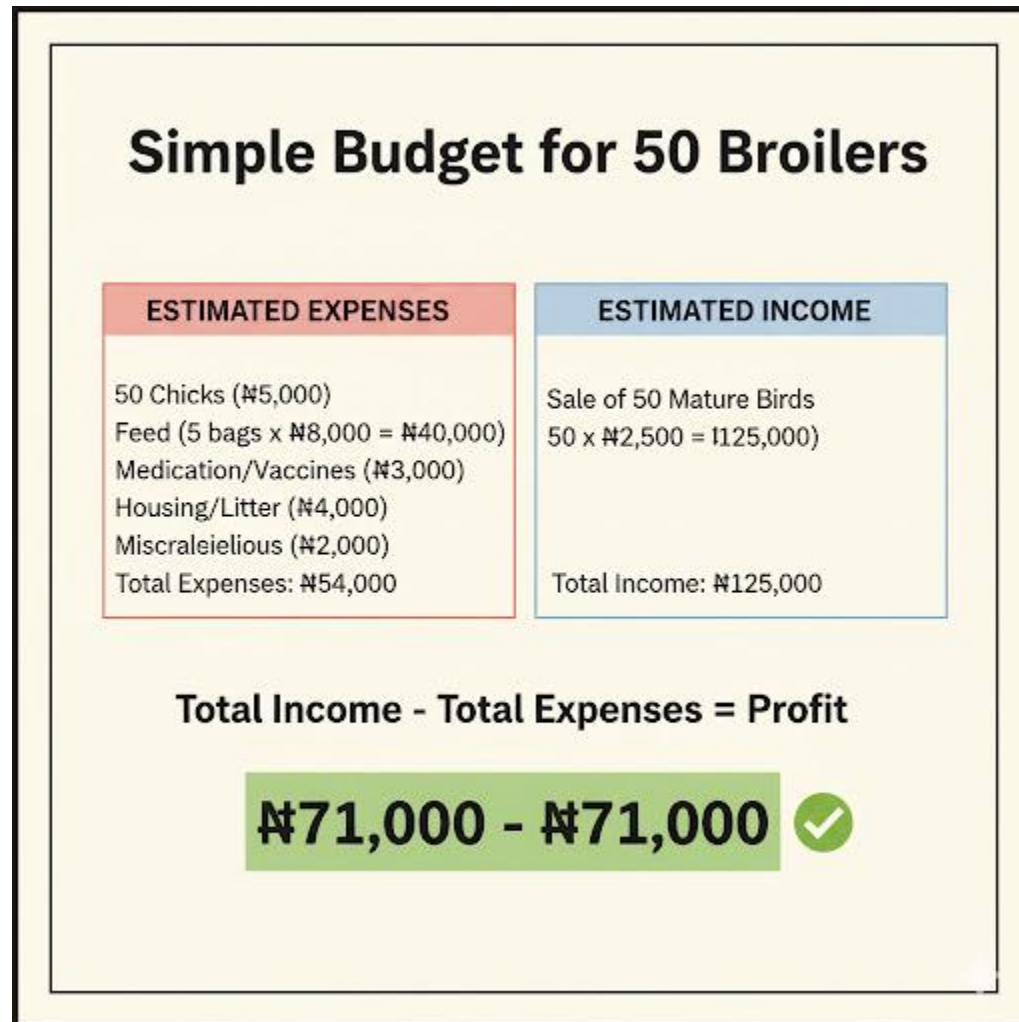


WEEK 10: PRACTICAL: BUDGETING IN LIVESTOCK FARMING

Topic: Practical: Budgeting in Livestock Farming

Content/Subtopics:

- Simple record-keeping and cost management exercises



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "budget" and "budgeting."
2. Differentiate between "fixed costs" and "variable costs."

3. List the common income and expense items for a small poultry farm.
4. Create a simple, one-cycle budget for a small livestock enterprise.
5. Explain how a budget is used for cost management.

Lesson Content:

1. What is a Budget?

- **Definition:** A **budget** is a *financial plan* that estimates the *income* and *expenses* for a specific project or period of time (e.g., "Budget for one batch of 50 broilers").
- **Budgeting:** The *process* of creating this plan *before* you start.
- **Importance:**
 - It tells you how much **capital (starting money)** you need.
 - It helps you control your costs (cost management).
 - It helps you predict your profit or loss.
 - It is required for getting a bank loan.

2. Cost Management: Fixed vs. Variable Costs To manage costs, you must know what they are.

- **A. Fixed Costs (Capital Costs):**
 - Costs for items that *do not change* and are *re-usable* for many production cycles.
 - *Examples:* Cost of building the pen, cost of feeders, cost of waterers, cost of hutch, cost of wheelbarrow.
- **B. Variable Costs (Operating Costs):**
 - Costs that *change* with each new production cycle. You must buy them again every time.
 - *Examples:* Cost of day-old chicks, cost of feed, cost of vaccines/drugs, cost of wood shavings (litter).
- **Total Cost = Fixed Costs + Variable Costs**

3. Simple Record-Keeping (Review) A budget is a *plan for the future*. Your *financial records* (Week 9, Term 1) are the *record of the past*. You use your old records to make a better, more accurate budget for the future.

4. Practical: Simple Budget for 50 Broilers (0-6 Weeks) Let's create a *plan* (a budget). We will make *estimations*. (Note: Prices are for *example only*).

A. ESTIMATED EXPENSES (What we will spend)

- **i. Fixed Costs (We assume we already have the pen)**
- **ii. Variable Costs (For this one cycle):**
 - - 1. 50 Day-Old Chicks (at N800/chick) = N40,000
 - - 2. Feed (3 bags of mash at N15,000/bag) = N45,000
 - - 3. Vaccines & Drugs = N5,000
 - - 4. Litter (Wood Shavings) = N2,000
 - - 5. Transport & Other = N3,000
 - **TOTAL ESTIMATED EXPENSES = N95,000**

B. ESTIMATED INCOME (What we will earn)

- We *assume* a 4% mortality (2 birds die). We have 48 birds left to sell.
- We *assume* they reach an average weight of 2kg.
- We *assume* we can sell them at N2,000/kg.
- Total KGs = 48 birds * 2 kg/bird = 96 kg
- **TOTAL ESTIMATED INCOME = 96 kg * N2,000/kg = N192,000**

C. ESTIMATED PROFIT

- **Profit = Total Income - Total Expenses**
- Profit = N192,000 - N95,000

- **ESTIMATED PROFIT = N97,000**

5. Cost Management By creating this budget, the farmer knows they must have at least N95,000 *before* they start.

- If the farmer's *records* at the end show that "Feed" was actually N50,000 (not N45,000), they can manage this cost by:
 - Finding a cheaper feed supplier.
 - Formulating their own feed.
 - Reducing feed wastage (using better feeders).

Evaluation:

1. What is a "budget"?
2. What is the difference between a fixed cost and a variable cost?
3. Is a "day-old chick" a fixed or variable cost?
4. Is a "feeder" a fixed or variable cost?
5. What is the formula for calculating profit?

Assignment:

1. Define Budget, Fixed Cost, and Variable Cost.
2. Why is a budget important *before* starting a farm?
3. **Practical:** Create a simple budget for a **rabbit farmer**.
 - **Goal:** Buy 1 Buck and 2 Does.
 - **Expenses:**
 - 1 Buck (at N5,000) = ?
 - 2 Does (at N5,000 each) = ?
 - 3 Hutches (at N8,000 each) = ?
 - Feed for 1 month = N6,000
 - **Calculate the TOTAL START-UP COST (Total Expenses).**
4. How does a farmer use a budget for "cost management"?
5. What is the difference between a "budget" and a "financial record"? (Hint: future vs. past).

WEEK 11: REVISION

Topic: Revision

Content/Subtopics:

- Review of the term's work

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key terms from the term.
2. Compare and contrast the different feeding and grazing systems.
3. Explain the importance of disease control (vaccination, sanitation, quarantine).
4. Define entrepreneurship and list its characteristics.
5. Explain the functions of money and the basics of budgeting.

Lesson Content:

1. Summary of Term 2 Topics (Management & Business):

- **Week 1 (Sheep & Goats):**
 - They are **Ruminants** (4-chambered stomach).
 - Feeding: Extensive, Semi-Intensive, Intensive (Zero Grazing).
 - Care of young: **Colostrum** (first milk) and **Navel disinfection** (Iodine) are critical.
 - Practices: Disbudding (remove horns), Castration (remove testes).
- **Week 2 (Disease Control):**
 - **Prevention is better than cure.**
 - Methods: **Vaccination** (PPR, Newcastle), **Sanitation** (cleaning), **Isolation** (of sick), **Quarantine** (of new), **Deworming**, **Dipping**.
- **Week 3 (Pasture):**
 - Land with forages for grazing.
 - Types: Natural (wild) vs. Artificial (planted).
 - Grazing Systems: Continuous (bad) vs. **Rotational** (good).
 - Maintenance: Weed control, fertilizer, controlled grazing.

- **Week 4 (Housing):**
 - Importance: Protection from weather and theft.
 - Features: Ventilation, Drainage, Space.
 - Materials: Concrete floor (best), Iron roof, Wire mesh.
 - **Orientation:** Must be **East-West** to keep animals cool.
- **Week 6 (Entrepreneurship):**
 - A *person* who takes a *risk* to start a *business* for *profit*.
 - Characteristics: Risk-taker, Innovative, Persistent, Hardworking.
- **Week 8 (Importance of Entrepreneurship):**
 - **Creates jobs**, grows the economy (GDP), provides food, promotes self-reliance.
 - Opportunities exist in Production, Processing, and Support Services.
- **Week 9 (Money):**
 - Replaced **Barter** (which had problems like "double coincidence of wants").
 - Functions: Medium of exchange, Unit of account, Store of value.
- **Week 10 (Budgeting):**
 - A *financial plan* (estimated income and expenses).
 - Costs: **Fixed** (re-usable, e.g., pen) vs. **Variable** (used up, e.g., feed).
 - **Profit = Income - Expenses.**

2. Class Activity: Final Q&A (Teacher asks rapid-fire questions from the term to prepare for the final examination).

- "What is a ruminant?"
- "What is colostrum?"
- "What is the difference between isolation and quarantine?"
- "What is rotational grazing?"
- "What is the best orientation for a farm building in Nigeria?"
- "What is an entrepreneur?"
- "List 3 characteristics of an entrepreneur."
- "List 3 business opportunities in livestock."
- "What are the 4 functions of money?"
- "What is the difference between a fixed cost and a variable cost?"

Evaluation: This week's evaluation is the Second Term (Final) Examination.

Assignment:

1. Study all notes from the term.
2. Make a list of 10 key terms from this term and their definitions (e.g., Ruminant, Estrus, Rotational Grazing, Quarantine, Entrepreneur, GDP, Barter, Budget, Fixed Cost, Variable Cost).
3. Draw a simple diagram of a rotational grazing system with 4 paddocks.
4. Why must a farmer be a "risk-taker"?
5. Create a simple budget (Income, Expenses, Profit) for a small business you would like to start (e.g., selling 20 rabbits).



THIRD TERM - JSS 2 LIVESTOCK FARMING LESSON NOTES



WEEK 1: BUSINESS START-UP IN LIVESTOCK FARMING

Topic: Business Start-up in Livestock Farming

Content/Subtopics:

- Steps in starting a small poultry, rabbit, or goat business



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "business start-up" in the context of livestock farming.
2. List the seven key steps for starting a small livestock business.
3. Explain the importance of a "business plan," even for a small farm.
4. Differentiate between "sourcing capital" and "sourcing foundation stock."

5. Create a simple, hypothetical checklist for starting a small poultry farm.

Lesson Content:

1. What is a Business Start-up in Livestock?

- **Definition:** A business start-up is the complete process of moving from an *idea* (e.g., "I want to sell eggs") to a fully *operational* business that is ready to serve customers and make a profit.
- It is a planned, systematic process that requires careful thought and resources. It is the first step in becoming a livestock entrepreneur.

2. Key Steps in Starting a Small Livestock Business Starting a farm, even a small one (like 10 rabbits or 50 broilers), requires a plan.

- **Step 1: Idea Generation and Opportunity (The "Why"):**
 - Identify a *need* in your community. Don't just start a farm because you like it; start it because people need the product.
 - *Questions:* Is there a market for eggs? Do people in my area buy rabbit meat? Is there a boarding school nearby that needs a supplier?
 - *Example Idea:* "My community has many canteens but no reliable egg supplier. I will start a small layer farm."
- **Step 2: Business Planning (The "How"):**
 - A business plan is your *roadmap* or *blueprint*. It is a written document that outlines your goals and how you will achieve them.
 - **A simple plan must include:**
 - **The Goal:** (e.g., "To raise 100 broilers every 8 weeks").
 - **Required Inputs:** (e.g., "I will need a pen, 100 chicks, 15 bags of feed, vaccines...").
 - **Budget:** (How much will all this cost? Where will I get the money?).
 - **Marketing:** (Who will I sell to? How will they know I exist?).
- **Step 3: Sourcing Capital (The "Money"):**
 - **Capital** is the money needed to start and run the business.

- **Sources:**
 - **Personal Savings:** The best and safest source.
 - **Family and Friends:** Loans or gifts from relatives.
 - **Bank Loans (Agri-loans):** Requires a very detailed business plan.
 - **Grants/Government Programs:** (e.g., N-Agro, Bank of Agriculture).
- *Note:* You need *Fixed Capital* (for pen, feeders) and *Working Capital* (for feed, chicks).
- **Step 4: Site Selection and Housing (The "Place"):**
 - (See Term 2, Week 4).
 - Choose a *location* (siting) that is secure, well-drained, and away from the family house (due to smell/fly).
 - *Construct* the appropriate housing (poultry pen, rabbit hutches) that is ventilated, secure, and easy to clean.
- **Step 5: Sourcing Foundation Stock (The "Animals"):**
 - This is a *critical* step. Your entire business depends on the quality of your first animals.
 - **"Foundation Stock"** is the first set of animals you buy for breeding or raising.
 - **Rule:** You *must* buy from a reputable hatchery (for chicks) or a known, healthy farm (for rabbits/goats). Do not buy from an open market where sick animals are mixed.
 - *Example:* For poultry, buy healthy Day-Old Chicks (DOCs). For goats, buy a good, healthy breed like the West African Dwarf.
- **Step 6: Sourcing Inputs and Equipment (The "Tools"):**
 - Before the animals arrive, you must have *everything* ready.
 - *Equipment:* Feeders, waterers, brooder (for chicks), nesting boxes (for rabbits).
 - *Inputs:* All the feed, vaccines, drugs, and litter (wood shavings) you will need for at least the first few weeks.
- **Step 7: Start Operations and Marketing:**
 - This is "Day 1." The animals arrive.
 - You must *immediately* start your biosecurity, feeding, and record-keeping.

- You must also start *marketing* (telling people what you have) *before* the animals are even ready (e.g., "My broilers will be ready in 6 weeks!").

Evaluation:

1. What is a "business start-up"?
2. What is the very first step in starting a livestock business? (Idea/Opportunity).
3. Why is a "business plan" important?
4. What is "foundation stock," and why is it important to buy from a reputable farm?
5. List the 7 key steps in starting a livestock business.

Assignment:

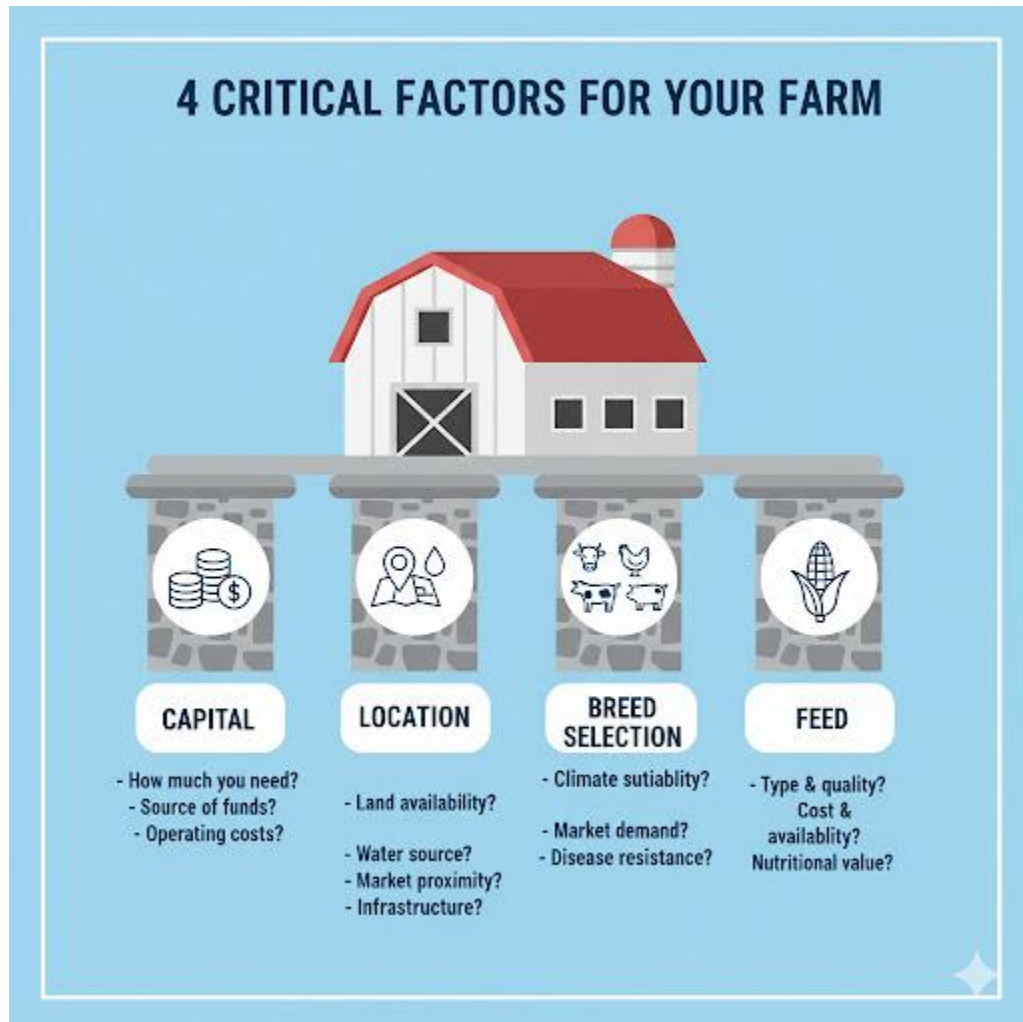
1. Define "Capital" and "Business Plan."
2. What is the difference between "Fixed Capital" and "Working Capital"? Give one example of each for a rabbit farm.
3. Why is it a bad idea to buy your foundation stock from an open market?
4. Imagine you want to start a 50-broiler business. Write a simple list of 5 "inputs" and 3 "equipment" items you must buy.
5. Of the 7 steps, which one do you think is the *most* difficult, and why?

WEEK 2: FACTORS TO CONSIDER BEFORE STARTING

Topic: Factors to Consider Before Starting

Content/Subtopics:

- Capital, location, breed selection, and feeding requirements



Performance Objectives: By the end of this lesson, students should be able to:

1. Identify and explain four critical factors to analyze before starting a livestock farm.
2. Explain the role and sources of capital (fixed and working).

3. List four key considerations for selecting a good farm location.
4. Explain the importance of selecting the right breed for the intended purpose.
5. Analyze the impact of feeding requirements on the business.

Lesson Content:

Before you spend one Naira, you must do a "feasibility study" by considering these critical factors. Failure to plan for these is the main reason new farms fail.

1. Capital (The Money)

- **Definition:** The total amount of money needed to start and run the business until it can pay for itself.
- **A. Fixed Capital:** Money for *re-usable* items (assets).
 - *Examples:* Cost of land, building the pen, buying feeders/waterers, digging a well, buying a wheelbarrow.
- **B. Working Capital (Operating Capital):** Money for *day-to-day* operations (consumables).
 - *Examples:* Money to buy chicks/kits, feed, vaccines, drugs, staff salaries, transportation.
- **Consideration:**
 - Do I have *enough* capital? A common mistake is spending all the money on the building and having *no money left* to buy feed (Working Capital).
 - *Source:* Where will I get the money? (Personal savings, loan, etc.).

2. Location (The "Siting")

- (See Term 2, Week 4). The location of the farm is critical.
- **Considerations:**
 1. **Accessibility:** Is there a good road to the farm? (To bring in feed and transport products out).
 2. **Proximity to Market:** Being close to your customers (e.g., a city, a market) reduces transport costs and stress on the animals.

3. **Water Supply:** Is there a reliable, clean source of water (borehole, well)?
Livestock (especially poultry) need a lot of water.
4. **Drainage:** Is the land well-drained? (Do not build in a swampy, flood-prone area).
5. **Biosecurity/Zoning:** The farm should be *away* from residential areas (to avoid complaints about smell/fly) and *away* from other livestock farms (to prevent disease spread).

3. Breed Selection (The "Animal")

- **Definition:** Choosing the specific *type* of animal that matches your business goal.
- **Considerations:**
 1. **Purpose (The Goal):** Why are you raising the animal?
 - *Poultry:* For meat (choose Broilers, e.g., Ross, Cobb). For eggs (choose Layers, e.g., Isa Brown). For both (choose Dual-Purpose, e.g., Kuroiler, Sasso).
 - *Goats:* For meat (choose West African Dwarf). For milk (choose a dairy goat, e.g., Saanen).
 2. **Hardiness (Adaptation):** Can the breed survive well in your local climate?
 - *Local breeds* (e.g., West African Dwarf goat) are very hardy, disease-resistant, and can survive on local forages.
 - *Exotic breeds* (e.g., Boer goat) grow *much* faster but are very expensive and require high-quality feed and veterinary care.
 3. **Reproductive Rate:** How fast do they multiply? (Rabbits multiply very fast, which is an advantage).
 4. **Availability:** Is the breed easy to find and buy in your area?

4. Feeding Requirements (The "Fuel")

- **Feed is the single biggest (60-70%) operating cost in livestock farming.**
- **Considerations:**
 1. **Availability:** Are the ingredients (maize, soya) or the finished feed readily available in your area, all year round?

2. **Cost:** What is the *price* of feed? If the price of maize is too high, your business may not be profitable.
3. **Quality:** Is the feed of good quality, or is it adulterated (mixed with chaff, etc.)?
4. **Storage:** Do you have a clean, dry, rodent-proof place to store the feed? (A feed store).

Evaluation:

1. What is the difference between "fixed capital" and "working capital"?
2. List three important factors to consider when choosing a *location* for a farm.
3. Why is "breed selection" important?
4. If your goal is to produce eggs, what type of chicken breed would you choose?
5. What is the single biggest *operating cost* in most livestock farms?

Assignment:

1. Define Fixed Capital and give two examples for a goat farm.
2. Define Working Capital and give two examples for a goat farm.
3. Explain the difference between a "local breed" and an "exotic breed" in terms of hardiness and cost.
4. You want to start a small rabbitry. List the four (4) main factors (Capital, Location, Breed, Feed) and write *one* important question you must answer for each. (e.g., For Location: "Is there a good water source?").
5. Why is it bad to build a farm in a swampy, flood-prone area?

WEEK 3: BUSINESS RECORDS

Topic: Business Records **Content/Subtopics:**

- Daily and monthly records
- Expenditure, income, and profit tracking

DAILY FINANCIAL RECORD SHEET

DAILY EXPENSES

Date		Item	Quantity	Unit Cost	Total Cost

DAILY INCOME

Date	Item Sold	Quantity	Unit Price	Total Price	Total Income

MONTHLY PROFIT CALCULATION

Total Income: _____
Total Expenses: _____
Net Profit: _____

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "business records" (financial records).
2. Differentiate between an "expenditure record" and an "income record."
3. Design a simple template for tracking daily/monthly expenses.
4. Define "profit" and calculate it using a simple formula.

5. Explain *why* a farmer must track profit.

Lesson Content: (This builds on Term 1, Week 9 and Term 2, Week 10)

1. What are Business Records?

- **Definition:** Business records (or **Financial Records**) are the systematic documentation of all *money* transactions related to the farm. They track every Kobo that comes *in* (Income) and every Kobo that goes *out* (Expenditure).
- **Record vs. Budget:**
 - A **Budget** (Term 2, Week 10) is a *plan for the future*.
 - A **Record** is a *report of the past*.
- **Rule:** You make a *budget*, and then you use *records* to see if you followed your budget.

2. Types of Financial Records

- **A. Expenditure Record (Expense Sheet / Cash Book - "Out"):**
 - **Definition:** A detailed list of *all* money spent by the business. This should be recorded *daily*.

Date	Item Description	Quantity	Unit Cost	Total Cost
Oct 1	Broiler Feed	2 bags	₦15,000	₦30,000
Oct 1	Vaccine (Lasota)	1 vial	₦500	₦500
Oct 2	Transport to Market	1 trip	₦1,000	₦1,000
Oct 3	Wood Shavings	1 bag	₦1,500	₦1,500
Total				₦33,000

- **Template (Daily):**

- **B. Income Record (Sales Sheet / Cash Book - "In"):**

- **Definition:** A detailed list of *all* money received by the business from sales.

- **Template (Daily):**

Date	Item Sold	Quantity	Unit Price	Total Income
Oct 5	Crate of Eggs	10 crates	₦3,000	₦30,000
Oct 5	Live Broiler	2 birds	₦7,000	₦14,000
Oct 6	Rabbit	1 buck	₦5,000	₦5,000
Total				₦49,000

3. Profit Tracking (Monthly Summary)

- At the end of every month (or production cycle), you must *track your profit*.
- **Step 1:** Calculate **Total Expenditure (TE)** from your expense sheet.
- **Step 2:** Calculate **Total Income (TI)** from your sales sheet.
- **Step 3:** Calculate **Profit or Loss**.
- **Formula:**
 - **Profit = Total Income - Total Expenditure**
 - (TI - TE = Profit)
- **Example (Monthly Summary):**
 - Total Income (from all sales in October) = N120,000
 - Total Expenditure (from all purchases in October) = N85,000
 - Profit = N120,000 - N85,000 = **N35,000 (Profit)**
- **If:**
 - Total Income = N120,000
 - Total Expenditure = N130,000

- Profit = N120,000 - N130,000 = **-N10,000 (Loss)**

4. Importance of Profit Tracking

- It is the *only* way to know if your business is succeeding or failing.
- It helps you identify *where* your money is going (e.g., "Wow, I spend 70% of my money on feed!").
- It helps you make decisions (e.g., "My rabbit business made a profit, but my goat business made a loss. I should focus on rabbits.").

Evaluation:

1. What is the difference between a "budget" and a "record"?
2. What is an "expenditure record"?
3. What is an "income record"?
4. What is the formula for calculating profit?
5. A farmer's Total Income is N200,000. His Total Expenditure is N150,000. What is his profit?

Assignment:

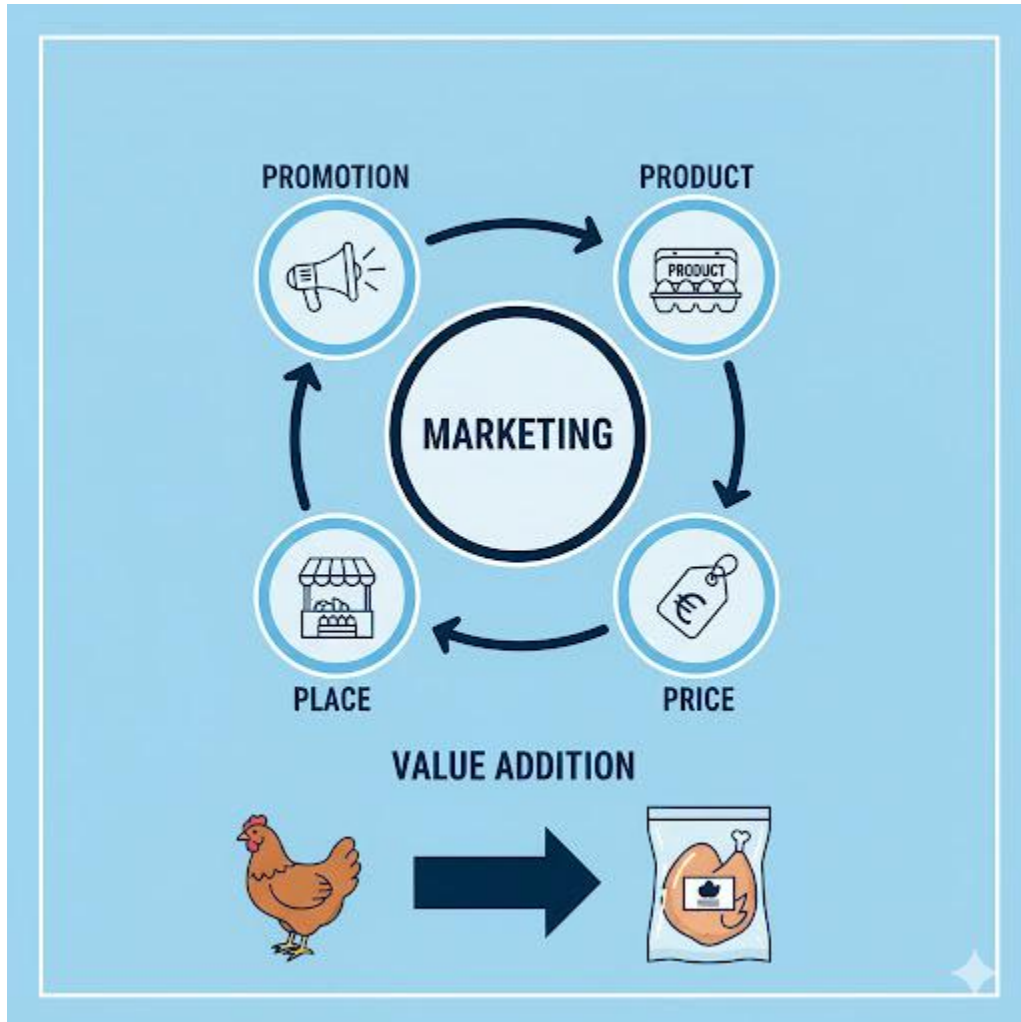
1. Define Expenditure, Income, and Profit.
2. Why is it important to track profit *monthly*?
3. Draw a simple, blank template for a "Daily Expenditure Record" (it must have 4 columns).
4. A rabbit farmer has the following for the month of May:
 - **Income:** Sold 10 rabbits for N4,000 each.
 - **Expenditure:** Bought feed for N15,000. Paid for transport for N3,000.
 - Calculate his (a) Total Income, (b) Total Expenditure, and (c) Total Profit for May.
5. What does it mean if a business makes a "loss"?

WEEK 4: MARKETING OF LIVESTOCK PRODUCTS

Topic: Marketing of Livestock Products

Content/Subtopics:

- Meaning and importance of marketing livestock products



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "marketing" and differentiate it from "selling."
2. Explain why marketing is essential for a livestock farmer.

3. List and explain the "4 Ps" of marketing.
4. Identify different "target markets" for a livestock product.
5. Explain how "value addition" is a marketing strategy.

Lesson Content:

1. What is Marketing? (Marketing vs. Selling)

- **Selling:** This is the *last step*. It is the simple act of exchanging a product (e.g., one chicken) for money.
- **Marketing:** This is the *entire process* of getting a product from the producer (farmer) to the consumer. It is a big, strategic plan.
- **Definition:** Marketing includes *all* the activities involved in:
 - **Researching** what the customer wants.
 - **Producing** a product that meets that want.
 - **Pricing** the product correctly.
 - **Promoting** (advertising) the product.
 - **Distributing** (placing) the product where the customer can buy it.
 - **Selling** the product.
- **Analogy:** Selling is just the *finish line*. Marketing is *planning and running the entire race*.

2. Importance of Marketing in Livestock Farming You can be the best farmer in the world, but if you cannot *market* your products, you will fail.

1. **It Creates Demand:** Marketing (like an advert) tells people *about* your product and makes them *want* to buy it.
2. **It Identifies Customers:** Marketing research helps you find your "target market" (who will buy your product?).
 - *Target Market (Eggs):* Households, schools, bakers, restaurants.
 - *Target Market (Goats):* Individuals (for festivals), restaurants, bukas.
3. **It Builds a Brand:** Good marketing builds a good reputation (e.g., "Mr. John's farm always has the freshest eggs"). This allows you to charge a better price.

4. **It Prevents Waste:** Marketing helps you plan your sales so you don't have 500 broilers ready with no buyers (they will keep eating your feed and reduce your profit).
5. **It Adds Value:** Marketing includes processing and packaging, which increases profit.

3. The "4 Ps" of Marketing (The Marketing Mix) This is the "recipe" for a successful marketing plan.

- **1. PRODUCT:**

- *What are you selling?* It must be a *quality* product.
- *Value Addition:* How can you make your product better than others?
 - *Raw Product:* Live chicken.
 - *Processed Product (Value added):* Dressed, frozen chicken.
 - *Packaged Product (Value added):* Graded, clean eggs in a branded carton.

- **2. PRICE:**

- *How much will it cost?* (See Term 3, Week 3 - Budgeting).
- The price must be *competitive* (similar to other sellers) but must also cover your *costs* and give you a *profit*.

- **3. PLACE (Distribution):**

- *Where will the customer buy it?*
- *Farm Gate:* The customer comes to your farm (easiest, but few customers).
- *Local Market:* You take the products to a stall.
- *Wholesale:* You sell in bulk to middlemen.
- *Contract/Delivery:* You deliver directly to hotels, schools, restaurants.

- **4. PROMOTION:**

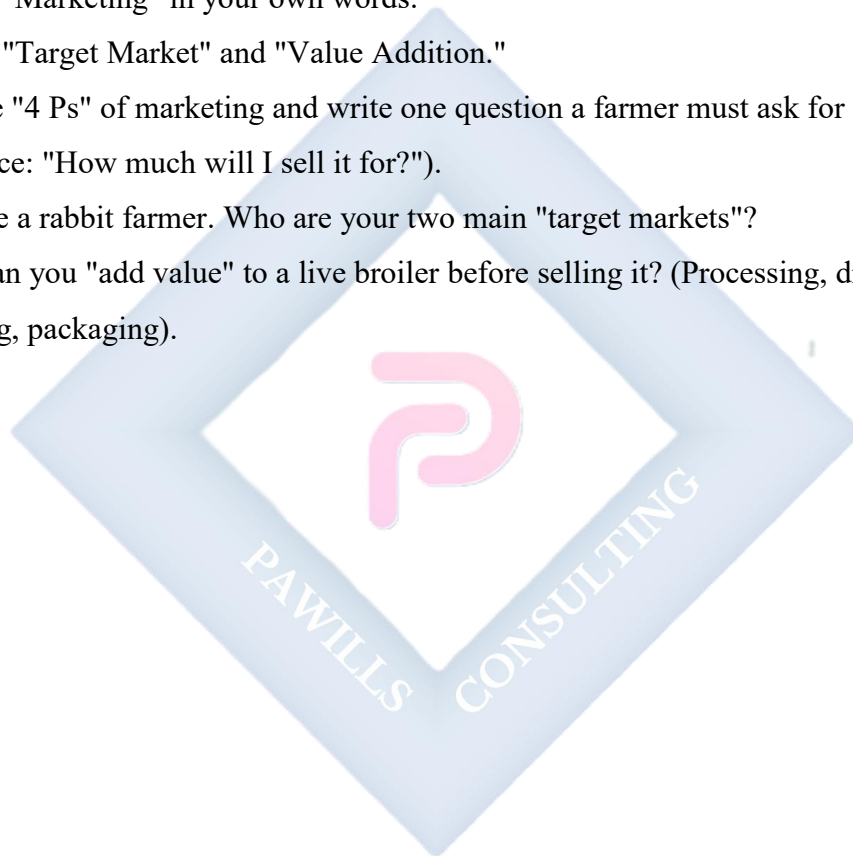
- *How will you tell people about your product?*
- *Word-of-Mouth:* Using your good reputation.
- *Advertising:* A sign at your farm gate, social media (WhatsApp, Instagram), business cards.
- *Branding:* Using professional packaging, labels, etc.

Evaluation:

1. What is the difference between "marketing" and "selling"?
2. State two reasons why marketing is important for a livestock farmer.
3. What are the "4 Ps" of marketing?
4. What is a "target market"? Give one example for a goat farmer.
5. What is "value addition"? Give one example.

Assignment:

1. Define "Marketing" in your own words.
2. Define "Target Market" and "Value Addition."
3. List the "4 Ps" of marketing and write one question a farmer must ask for each "P." (e.g., For Price: "How much will I sell it for?").
4. You are a rabbit farmer. Who are your two main "target markets"?
5. How can you "add value" to a live broiler before selling it? (Processing, dressing, freezing, packaging).



WEEK 6: MARKETING CHANNELS

Topic: Marketing Channels

Content/Subtopics:

- From producer to consumer
- Role of middlemen and wholesalers



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "marketing channel" (distribution channel).
2. Draw a diagram of a simple, two-level, and three-level channel.
3. Define "middleman," "wholesaler," and "retailer."

4. Explain the *function* (role) of middlemen/wholesalers.
5. Analyze the advantages and disadvantages of using middlemen.

Lesson Content:

1. What is a Marketing Channel?

- **Definition:** A marketing channel (or "distribution channel") is the *path* or *route* a product takes to get from the **Producer** (the farmer) to the final **Consumer** (the person who eats it).
- It answers the "PLACE" question from the 4 Ps.

2. Types of Marketing Channels

- **A. Channel 1: Direct Channel (Zero-Level)**
 - **Path:** Producer -> Consumer
 - **Description:** The farmer sells *directly* to the public.
 - **Examples:** Selling eggs at your farm gate, a farmer having a stall in the local market.
 - **Advantages:** Farmer gets 100% of the profit (no middlemen).
 - **Disadvantages:** Can only sell in small quantities; takes a lot of the farmer's time.
- **B. Channel 2: Retailer Channel (One-Level)**
 - **Path:** Producer -> Retailer -> Consumer
 - **Description:** The farmer sells to a shop, restaurant, or hotel (the retailer), who then sells to the public.
 - **Example:** Your farm supplies eggs to 10 local supermarkets.
 - **Advantages:** Farmer can sell larger quantities; the retailer handles the marketing.
 - **Disadvantages:** The retailer takes a share of the profit (buys at a lower price).
- **C. Channel 3: Wholesaler Channel (Two-Level)**
 - **Path:** Producer -> Wholesaler -> Retailer -> Consumer
 - **Description:** This is for *very large* farms. The farmer sells *all* his products (e.g., a truckload of 5,000 broilers) to *one* big wholesaler (distributor).

- The wholesaler then "breaks the bulk" and sells in smaller quantities (e.g., 50 chickens) to many different retailers (market women, shops).
- **Advantages:** Farmer sells everything at once; simplest solution for a large farm.
- **Disadvantages:** Farmer gets the *lowest* price per item, as both the wholesaler and retailer must add their profit.

3. Roles of Middlemen (Wholesalers and Retailers)

- **Definition:** A "middleman" is anyone in the marketing channel between the producer and the consumer (e.g., wholesaler, agent, retailer).
- **Their Functions (Why they are important):**
 1. **Bulk-Breaking:** They buy a large quantity (a truckload) and break it into small quantities that consumers want (one chicken).
 2. **Storage/Warehousing:** They often have storage (cold rooms, warehouses) to hold the products, taking that risk away from the farmer.
 3. **Financing:** They often pay the farmer *in cash* immediately, while the farmer might have to wait for consumers to pay.
 4. **Transportation:** They provide the trucks and logistics to move the goods from the farm to the city.
 5. **Market Information:** They tell the farmer what the market *wants* (e.g., "Customers want bigger chickens this month").
- **Disadvantages of Middlemen:** They *add cost*. Every middleman adds their own profit, which makes the final product more expensive for the consumer.

Evaluation:

1. What is a "marketing channel"?
2. What is the *shortest* possible channel? (Producer -> Consumer).
3. What is the main difference between a "wholesaler" and a "retailer"?
4. What does "bulk-breaking" mean?
5. List three important functions (roles) of middlemen.

Assignment:

1. Define "Marketing Channel," "Wholesaler," and "Retailer."
2. Draw a simple diagram showing a One-Level Channel and a Two-Level Channel.
3. What is one advantage of a farmer using a *direct* (Zero-Level) channel?
4. What is one advantage of a farmer using a *wholesaler* (Two-Level) channel?
5. Some people say middlemen are "bad." Explain one *good* reason why they are important.

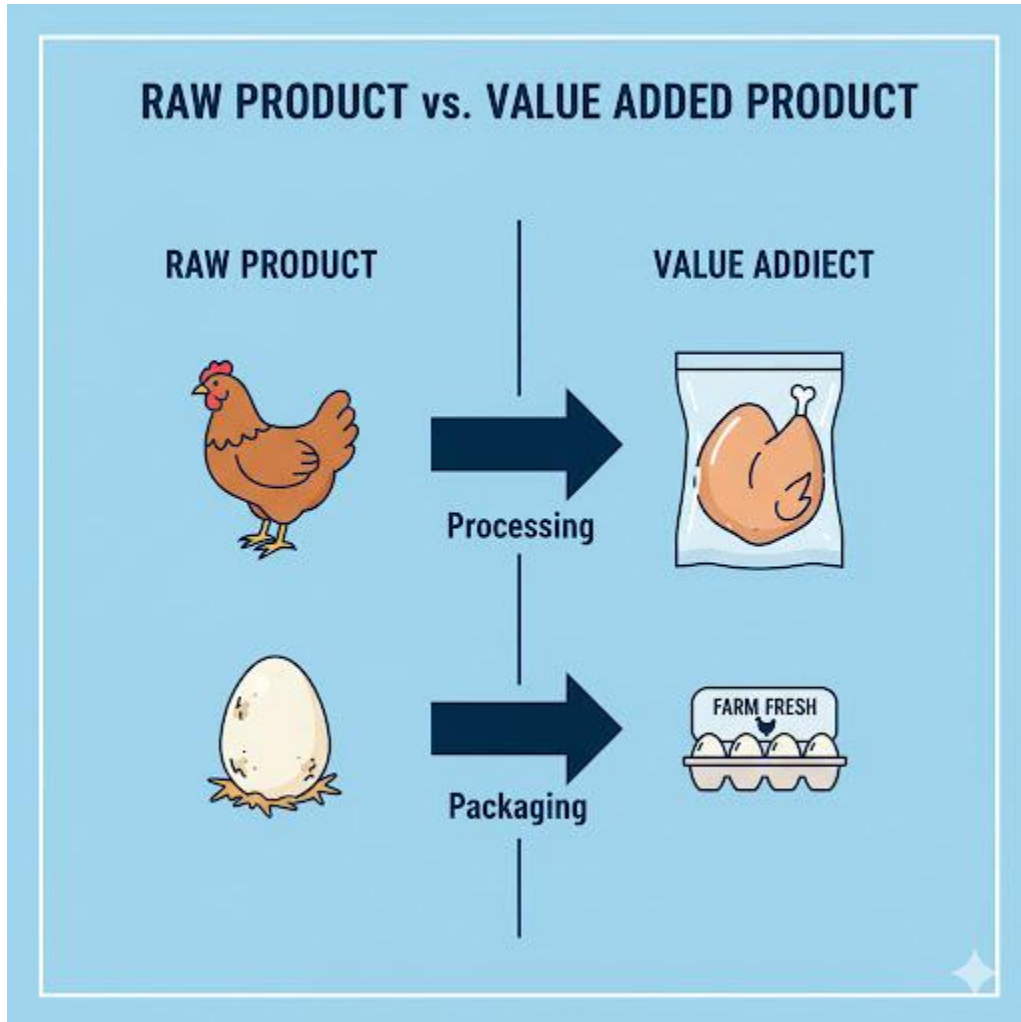


WEEK 8: PACKAGING AND PROCESSING

Topic: Packaging and Processing

Content/Subtopics:

- Methods of packaging eggs, meat, hides, and fur for sale



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "processing" (value addition) and "packaging."
2. State three key reasons for processing livestock products.
3. State three key reasons for packaging livestock products.
4. Describe the correct packaging methods for eggs.

5. Describe the processing and packaging methods for meat and hides/skin.

Lesson Content:

1. Definitions

- **Processing (Value Addition):**

- The act of *transforming* a raw farm product (like a live goat) into a different, more usable, or more valuable form (like goat meat, or leather).
- **Reasons for Processing:**
 1. **Preservation:** To make the product last longer (e.g., smoking fish, freezing chicken).
 2. **Convenience:** To make it easier for the consumer (e.g., selling "dressed" chicken).
 3. **To Add Value:** Processed products sell for a *higher price* than raw products (e.g., sausages cost more than raw pork).

- **Packaging:**

- The act of *enclosing* or *protecting* the product in a container (e.g., a crate, a bag, a carton).
- **Reasons for Packaging:**
 1. **Protection:** To protect the product from breaking (eggs), spoiling, or getting dirty.
 2. **Convenience:** To make it easy to handle, transport, and store (e.g., a crate of eggs).
 3. **Promotion (Branding):** To make the product attractive and provide information (e.g., a branded egg carton with your farm's name and an expiry date).

2. Packaging and Processing of Specific Products

- **A. Eggs:**

- **Processing:**
 - **Cleaning:** Dry-cleaning with sandpaper or a cloth.

- **Candling:** (Term 1, Week 4) Shining a light through to check for cracks, freshness, and blood spots.
 - **Grading:** Sorting the eggs by size/weight (e.g., Small, Medium, Large).
- **Packaging:**
 - **Bulk:** Eggs are placed (pointed end down) into paper or plastic **crates** (usually holding 30 eggs).
 - **Retail:** Eggs are packed into branded **cartons** (holding 6, 10, or 12 eggs) for sale in supermarkets.
- **B. Meat (Poultry, Rabbit, Goat):**
 - **Processing:**
 - **Slaughtering:** (Term 1, Week 4) Humane bleeding.
 - **Dressing:** (Scalding, de-feathering/flaying, evisceration).
 - **Portioning:** Cutting the dressed carcass into specific parts (e.g., drumsticks, wings, breast).
 - **Preservation:**
 - **Freezing/Refrigeration:** The most common method.
 - **Smoking:** Using wood smoke to dry and preserve the meat (common for fish/pork).
 - **Salting/Curing:** Using salt to draw out moisture.
 - **Packaging:**
 - **Fresh Meat:** Placed on a foam tray and wrapped in plastic **cling film**.
 - **Frozen Meat:** Packed into sealed, airtight **plastic bags** and stored in a deep freezer.
 - **Smoked/Dried Meat:** Wrapped in paper or packed in bags.
- **C. Hides (from Cattle) and Skin (from Goats/Sheep):**
 - **Processing:**
 - **Flaying:** The skilled act of removing the skin from the carcass without making "flay cuts" (holes).
 - **Curing (Preservation):** The skin must be preserved *immediately* or it will rot.

- **Method 1 (Salting):** Rubbing large amounts of salt all over the flesh side.
- **Method 2 (Drying):** Stretching the skin on a frame in the sun.
- **Packaging:**
 - After curing, the hard, dry skins are folded neatly, bundled, and tied with rope for transport to a **tannery** (where they are made into leather).

Evaluation:

1. What is "processing," and what is "packaging"?
2. State two reasons *why* we process livestock products.
3. State two reasons *why* we package livestock products.
4. What is the best way to package eggs for (a) bulk transport and (b) retail sale?
5. What is "flaying," and what is "curing" in hide processing?

Assignment:

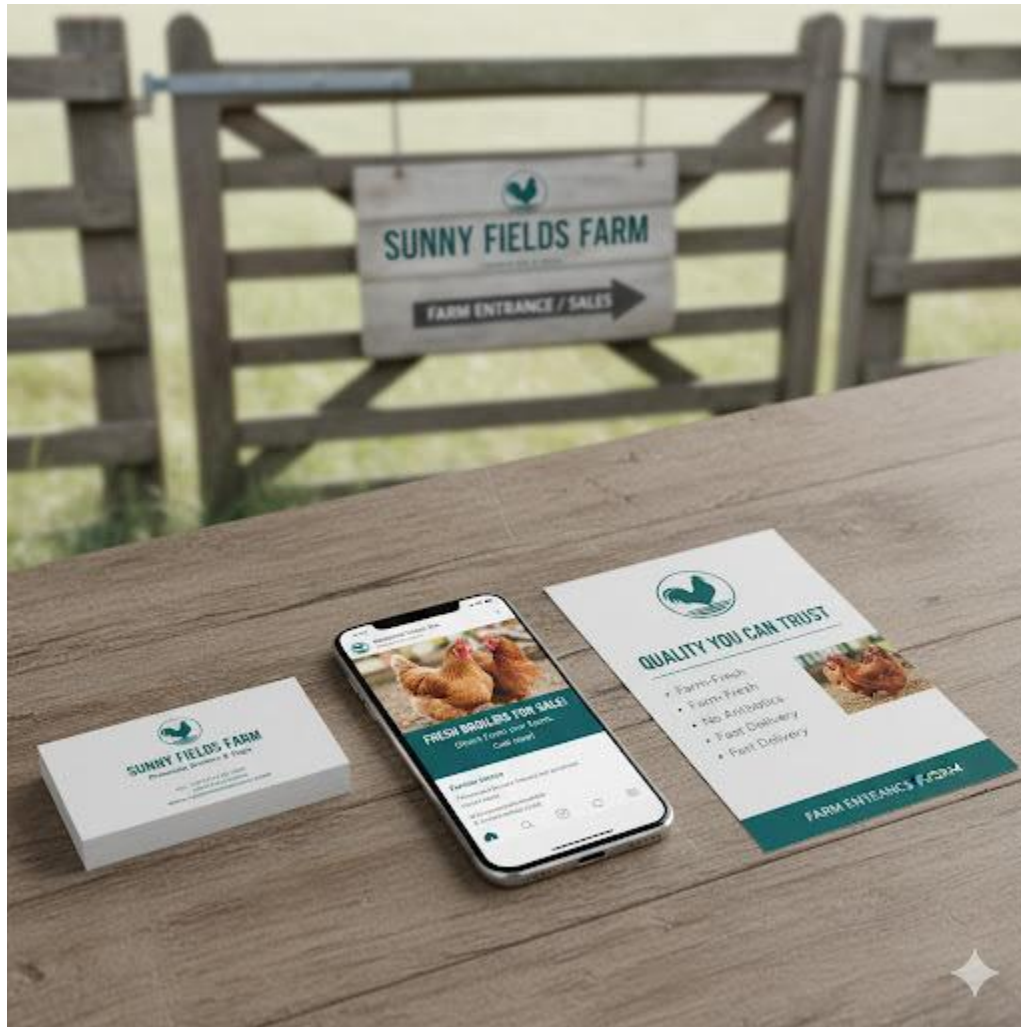
1. Define "Processing" and "Packaging."
2. What is the difference between "candling" and "grading" eggs?
3. List the steps for processing a live chicken into a frozen, dressed bird.
4. Why is it important to cure a goat's skin after flaying?
5. (Practical) Go to a local shop or supermarket. Find one livestock product (e.g., eggs, frozen chicken, milk). Write down (a) the product, (b) the packaging material, and (c) the brand name on the package.

WEEK 9: MARKETING POULTRY, RABBIT, SHEEP, AND GOAT PRODUCTS

Topic: Marketing Poultry, Rabbit, Sheep, and Goat Products

Content/Subtopics:

- Sales, branding, and advertising techniques



Performance Objectives: By the end of this lesson, students should be able to:

1. Define "branding" and "advertising."
2. Differentiate between branding and advertising.
3. List three specific sales techniques for livestock products.
4. Explain the role of social media in modern livestock marketing.

5. Develop a simple advertising message for a livestock product.

Lesson Content:

This lesson combines all our marketing topics into a practical strategy.

1. Branding (Creating Your Identity)

- **Definition:** Branding is the process of creating a unique, professional, and memorable **identity** for your farm. It's the "personality" of your business.
- **Goal:** To make a customer choose *your* chicken over all the other chickens in the market.
- **Techniques:**
 - **A Good Name:** A catchy, trustworthy name (e.g., "Excel Farms," "Joyful Eggs").
 - **A Logo:** A simple, professional symbol for your farm.
 - **Professional Packaging:** (Week 8) Using branded bags or egg cartons.
 - **Labels:** Putting a small brand label on your dressed chicken.
 - **Consistency:** Always selling high-quality products. Your *quality* is your best brand.

2. Advertising (Telling People You Exist)

- **Definition:** Advertising is the *action* of sending a message to the public to *promote* your product. It is one part of "Promotion" (in the 4 Ps).
- **Branding vs. Advertising:**
 - **Branding** is *who you are*.
 - **Advertising** is *what you say*.
- **Techniques:**
 - **Farm Gate Sign:** A large, clear sign: "EXCEL FARMS: Fresh Eggs and Broilers Sold Here. Call 080-..."
 - **Business Cards/Fliers:** Simple cards to give to potential customers.
 - **Social Media (WhatsApp/Instagram/Facebook):**
 - This is the cheapest and most powerful tool today.
 - Post *pictures* and *videos* of your healthy animals.

- Post *customer testimonials* (a happy customer's review).
- Post *adverts* (e.g., "Sallah Rams for sale! Call now!" "Fresh broilers, N6,000. Free delivery in this area.").
- **Word-of-Mouth:** Telling your friends, family, and neighbours to tell *their* friends.

3. Sales Techniques (Making the Sale)

- **A. Direct Sales (Farm Gate / Market Stall):**
 - **Technique:** Build a good relationship with customers. Be polite, honest, and reliable. (Good Customer Service).
- **B. Contract Sales (Business-to-Business, B2B):**
 - **Technique:** Identify businesses that *need* your product.
 - *Action:* Write a simple "Proposal Letter" to local hotels, restaurants, boarding schools, and supermarkets. Offer to be their regular, reliable supplier.
- **C. Promotional Sales:**
 - **Technique:** Offering special deals to attract new customers.
 - *Examples:* "Buy 5 crates of eggs, get 10% off." "Sallah Promo: Book your ram early!" "Refer a friend and get a free chicken."
- **D. Festive Sales:**
 - **Technique:** Plan your production to match *festivals*.
 - *Action:* Raise your broilers/goats to be ready and at their biggest *just before* Christmas, Easter, or Sallah, when demand and prices are highest.

Evaluation:

1. What is "branding"?
2. What is "advertising"?
3. What is the difference between branding and advertising?
4. List two advertising techniques a farmer can use.
5. What is a "contract sale"?

Assignment:

1. Define Branding, Advertising, and Sales Promotion.
2. You are starting a new poultry farm called "Speedy Broilers." Write a short (15-second) radio jingle or a WhatsApp advert for your business.
3. What is the "festive sales" technique, and why is it profitable?
4. Why is good customer service a sales technique?
5. (Project) Design a simple *logo* for your imaginary farm.



WEEK 10: PROJECT WORK

Topic: Project Work

Content/Subtopics:

- Practical farm visit or market survey

MARKET SURVEY REPORT

Group Members: _____
Date: _____
Market Visited: _____
Products Researched: _____

Questions We Asked	What We Found Out
• What products are most popular? • What are current market prices? • What do customers want?	

CONCLUSION

Performance Objectives: By the end of this lesson (project), students should be able to:

1. Work collaboratively in a group to conduct a field survey.
2. Design a simple questionnaire for a farmer or marketer.
3. Practice communication skills by interviewing a farmer/marketer.
4. Collect and record real-world data (prices, channels, challenges).

5. Write a simple, one-page report summarizing their findings.

Lesson Content:

This project is a practical application of the entire term's work on the "business" of livestock.

Project: A Market Survey of Livestock Products (A farm visit is ideal but harder to organize. A market survey is highly practical.)

- **Goal:** To understand the *real-world* marketing channels, prices, and challenges in the local livestock market.
- **Groups:** Students will be divided into groups of 4-5.
- **Location:** A local market (the section for live animals, or the frozen food section).
- **Task:** Each group must go to the market (as homework or a planned trip) and interview at least *two* sellers (retailers or wholesalers).

1. Preparation (In Class): Design the Questionnaire Each group must prepare a short list of questions.

- **Sample Questions for a Live Chicken Seller:**

1. Good morning. We are students from [School Name]. We are studying livestock. Please, can we ask you 2-3 questions?
2. Where do you get your chickens from? (A farm? A big market?) [*This identifies the marketing channel*].
3. How much do you sell a live broiler for today? [*This identifies the price*].
4. Does the price change? When is it highest? [*Identifies festive sales*].
5. What do your customers look for when buying a chicken? (e.g., Big size? Yellow skin?) [*Identifies product quality*].
6. What is your biggest challenge in this business? (e.g., Transport? Chickens dying?) [*Identifies challenges*].

- **Sample Questions for an Egg Seller:**

1. ...
2. Where do you buy your eggs from? (A farm? A wholesaler?)

3. How much is one crate of eggs today?
4. What is the difference in price between the big eggs and the small eggs?
[Identifies grading].
5. What is your biggest challenge? (e.g., Breakage, transport).

2. Execution (The Field Trip):

- Students go to the market, politely conduct their interviews, and write down the answers.
- **Safety:** Students must be in groups, be respectful, and not disrupt business.

3. Presentation (The Report):

- Each group will write a simple, one-page report titled "A Market Survey of [Market Name]."
- **Report must include:**
 - **Introduction:** Where and when they went.
 - **Findings (Summary):** "We interviewed two egg sellers. We found that..." (Summarize their answers).
 - **Conclusion:** "We learned that the biggest challenge for sellers is transport and that prices are highest at Christmas."

Evaluation: This is a practical project. The final *Report* will be graded based on:

1. Clarity of the report (10 marks).
2. Quality of the questions asked (10 marks).
3. Detail of the findings (data collected) (20 marks).
4. Group collaboration (10 marks).

Assignment (The Project):

1. Form your groups and finalize your questionnaire (get it approved by the teacher).
2. Conduct your market survey (this is the homework for the week).
3. Write your group's one-page report.
4. Be prepared to present your findings to the class in a 3-minute oral presentation.



WEEK 11: REVISION

Topic: Revision

Content/Subtopics:

- Review of the term's topics and key concepts

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key business terms from the term.
2. Outline the steps for starting a livestock business.
3. Compare and contrast different marketing channels.
4. Explain the importance of processing, packaging, and branding.
5. Review the basics of profit calculation.

Lesson Content:

1. Summary of Term 3 Topics (The "Business & Marketing" Term):

- **Week 1 (Start-up):** We learned the 7 steps: Idea -> Business Plan -> Capital -> Location/Housing -> Foundation Stock -> Inputs/Equipment -> Start Operations.
- **Week 2 (Factors):** Before starting, you must consider 4 key factors: **Capital** (fixed/working), **Location** (access, water, drainage), **Breed Selection** (purpose, hardiness), and **Feed** (cost, availability).
- **Week 3 (Business Records):** Tracking money.
 - **Expenditure:** Money out.
 - **Income:** Money in.
 - **Profit = Total Income - Total Expenditure.**
- **Week 4 (Marketing):** The *entire process* of getting a product to the consumer.
 - **The 4 Ps:** Product (quality, value added), Price (cost + profit), Place (distribution), Promotion (advertising).
- **Week 6 (Marketing Channels):** The *path* from producer to consumer.
 - P -> C (Direct)

- P -> Retailer -> C (One-Level)
- P -> Wholesaler -> Retailer -> C (Two-Level)
- **Middlemen** (wholesalers, retailers) perform functions like bulk-breaking, storage, and transport.
- **Week 8 (Processing & Packaging):**
 - **Processing:** Adding value and preserving (e.g., dressing, freezing, smoking).
 - **Packaging:** Protecting and promoting (e.g., crates, cartons, plastic bags).
- **Week 9 (Marketing Techniques):**
 - **Branding:** Creating your *identity* (name, logo, labels).
 - **Advertising:** *Telling* people you exist (signs, social media).
 - **Sales:** Direct, Contract, Promotional, Festive.
- **Week 10 (Project):** We conducted a market survey to find real-world prices and channels.

2. Class Activity: Final Q&A (Teacher asks rapid-fire questions from the term to prepare for the final examination).

- "What is a business plan?"
- "What is the difference between fixed and working capital?"
- "What is the formula for profit?"
- "What are the 4 Ps of marketing?"
- "What is the difference between a wholesaler and a retailer?"
- "What does 'bulk-breaking' mean?"
- "What is 'value addition' in meat processing?"
- "What is the difference between 'branding' and 'advertising'?"

Evaluation: This week's evaluation is the Third Term (Final) Examination.

Assignment:

1. Study all notes from the term.

2. Make a list of 10 key terms from this term and their definitions (e.g., Business Plan, Capital, Expenditure, Profit, Marketing, Wholesaler, Retailer, Processing, Branding, Advertising).
3. Draw a diagram showing the three main marketing channels.
4. You are starting a farm. Create a (a) Business Name and (b) a simple Advert (like a WhatsApp broadcast message) for your farm.
5. A farmer's records show: Total Income = N300,000. Total Expenses = N220,000. What is the profit?

